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IR-IDS: A network intrusion detection method based on causal feature selection and explainable model optimization

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ABSTRACT

With the rapid advancement of computer network technologies, the complexity of cybersecurity issues has grown significantly. Intrusion Detection Systems (IDS), serving as the first line of defense against network attacks, are vital components in ensuring network security. However, traditional IDS often struggle to balance the robustness of detection capabilities with the interpretability of the model. To address these challenges, this paper proposes an interpretable and robust intrusion detection method (IR-IDS). The proposed approach begins by efficiently and accurately selecting the optimal feature subset for predicting the target variable, using a causal effect-based conditional testing method and a Markov blanket search algorithm. Subsequently, it enhances the decision tree algorithm using Shapley values, enabling fine-grained classification of attacks. Finally, by integrating Kolmogorov–Arnold Networks (KAN) and Conditional Variational Autoencoders (CVAE), the method further improves the detection of unknown attacks. Experimental results demonstrate that the proposed method outperforms existing techniques on five datasets, including CIC-IDS2017, CSE-CIC-IDS2018, CIC-DDoS2019, CIC-UNSW-NB15 and CIC-IoT-IDAD-2024, with multi-class accuracies of 98.83 %, 99.37 %, 99.57 %, 99.52 % and 97.11 %, respectively. From the results, it can be seen that this method not only ensures the interpretability of the model but also improves the accuracy and robustness of intrusion detection.

1. Introduction

With the rapid development and widespread adoption of computer network technologies, cybersecurity has become increasingly complex and critical. Network attack methods continue to evolve, posing significant challenges to the security of digital infrastructures (Viegas et al., 2021). As a result, safeguarding computer networks from such attacks has become a central concern in the field of information technology.

Intrusion Detection Systems (IDS) are a critical component of network security, acting as the first line of defense against network intrusions. An IDS monitors network traffic in real-time, analyzing packet contents to detect potential anomalous behavior (Ning et al., 2021). When malicious activity is detected, the IDS generates alerts, allowing network administrators to take timely action and implement protective measures.

Currently, a large number of papers have utilized feature selection methods to enhance the performance of IDS (Fatima et al., 2024; Mohiuddin et al., 2023). Feature selection methods are commonly classified into three main types: filter, embedded, and wrapper (Lyu et al., 2023). Filter-based methods perform feature selection independently, without relying on the learning process of the classifier itself. This independence enhances their compatibility and modularity, making them widely applicable. Filter-based methods typically consist of two key components: feature selection and classification algorithms. Feature selection involves identifying the most relevant features from large volumes of network data to improve attack detection accuracy while reducing computational complexity. However, existing feature selection methods often focus on data correlations and feature dependencies, neglecting the complex causal relationships between features (Wu et al., 2022). This limitation becomes apparent when processing high-dimensional network data, where redundant features can hinder the performance of classification algorithms (Ling et al., 2023). To address this issue, this paper introduces a causal-effect-based feature selection method that identifies the Markov blanket of the target

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0167-4048/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

variable, improving both the efficiency and accuracy of feature selection.

Classification algorithms are another crucial factor in the effectiveness of an IDS. Traditional algorithms, such as Decision Trees (DT), Support Vector Machines (SVM) (Jeong et al., 2022), and K-Nearest Neighbors (KNN) (Pajouh et al., 2019), can effectively classify anomalies. However, these methods primarily focus on detecting known attacks and often lack the robustness needed to identify unknown attacks. As a result, deep learning-based methods such as Convolutional Neural Networks (CNN) (Yue et al., 2022), Autoencoders (AE) (Megantara et al., 2021), and Long Short-Term Memory networks (LSTM) (Elsayed et al., 2022) have garnered increasing attention. These techniques can learn from vast amounts of network traffic data, uncovering attack patterns and identifying previously unseen attack types. However, these methods are often criticized for their lack of transparency, as they function as "black-box" models, making it difficult to interpret decision-making processes and provide users with clear insights.

To overcome these challenges, this paper proposes an interpretable and robust intrusion detection method (IR-IDS) that enables fine-grained classification of known attacks and anomaly detection for unknown attacks. IR-IDS consists of two main stages: causal feature selection and robust anomaly classification. These modules are independent and can be easily integrated with other classification or feature selection algorithms, offering excellent flexibility and compatibility.

The causal feature selection component uses conditional testing based on the causal effects of features on the target variable, selecting the optimal feature subset for predicting the target variable. This process, grounded in conditional tests and the Markov blanket principle, ensures both efficiency and accuracy in feature selection.

Next, based on the selected features, the decision tree model is optimized using Shapley values, enabling effective pre-classification of network traffic. The results are then passed to the Kolmogorov–Arnold Networks and Conditional Variational Autoencoder (KAN—CVAE) model, where reconstruction errors are used to detect anomalies associated with unknown attacks, enhancing the system's robustness. The inclusion of Kolmogorov–Arnold Networks (KAN) in the CVAE model significantly improves the interpretability of the overall system.

In summary, the key contributions of this paper include:

(1) A highly efficient Markov blanket search algorithm based on causal feature selection principles, which identifies the optimal feature subset for predicting the target variable with superior efficiency and accuracy.
(2) A conditional testing method based on propensity scores, which determines the causal dependence between features and the target variable, ensuring both interpretability and avoidance of dimensionality issues.
(3) An interpretable classification model, Shapley Tree, optimized with Shapley values, enabling pre-classification of attack traffic with fine-grained detection.
(4) The incorporation of KAN networks into the CVAE model to create KAN—CVAE, an interpretable encoding model that performs anomaly detection for unknown attacks by calculating reconstruction errors and applying Extreme Value Theory.

The structure of this paper is organized as follows: Section 2 presents a review of related work, Section 3 provides a detailed explanation of the IR-IDS method, Section 4 validates the performance of IR-IDS on five datasets, and Section 5 concludes the paper with a summary of the findings.

2. Related work

In recent years, various methods have been proposed for IDS to improve detection performance, adapt to new types of attacks, and handle complex network environments.

Traditional classification algorithms, due to their efficiency and simplicity, have been widely used in known attack scenarios. DAE (Megantara and Ahmad, 2021) proposed a feature selection and dimensionality reduction method that combines decision trees and Local Outlier Factor (LOF), optimizing the computational efficiency of network traffic detection. KNN (Pajouh et al., 2019) enhanced the detection performance for low-frequency attacks through two levels of dimensionality reduction and a dual-classification module. However, the method, which relies on PCA and LDA as dimensionality reduction tools, although improving performance, may result in the loss of some feature information. CSA-RF (Mohiuddin et al., 2023) combined the Enhanced Cuckoo Search Algorithm (CSA) with Particle Swarm Optimization (PSO) for feature selection and used Weighted Least Squares Random Forest (WLS-RF) for classification, improving detection performance. However, CSA and PSO are highly dependent on parameters such as convergence speed and step size, making the tuning process complex and time-consuming. SCAESVM (Wang et al., 2020) combined Stacked Contractive Autoencoders (SCAE) with Support Vector Machines (SVM), aiming to address the high dimensionality and redundancy of network traffic in cloud computing environments. SVM (Jeong et al., 2022) proposed an adaptive intrusion detection system to address CAN bus security threats, achieving high-precision malicious node detection through autocorrelation features and SVM, effectively countering the impact of temperature variations. However, this system relies on temperature sensor hardware and cannot adapt to environments without sensors. AGC (Wu et al., 2025) addresses the intrusion detection problem under limited labels by utilizing anchor graph clustering. It integrates the construction, learning, and clustering of anchor graphs into a unified framework, achieving fine-grained classification for low-quality labels. However, it also fails to account for the issue of unknown attack scenarios.

Although these methods excel in fine-grained classification of known attacks, they mainly rely on training with known categories, which limits their generalization ability and makes them less effective at detecting unknown attacks. This limitation significantly restricts their robustness and practical application in open environments.

To improve model generalization, deep learning-based methods have attracted considerable attention. CGAN (Wu et al., 2023) proposed an intelligent intrusion detection method that combines Fuzzy Rough Sets, Convolutional Neural Networks (CNN), and Generative Adversarial Networks (GAN). While it excels in feature extraction and classification performance, GANs are highly dependent on the quality of training data. If the input data contains noise, the generated data may introduce errors, reducing detection performance. CNN3D (Yue et al., 2022) tackles the issues of intra-class diversity and inter-class similarity by incorporating Contrastive Learning, which effectively reduces false positives and false negatives and performs well in detecting complex traffic and novel attacks. However, contrastive learning is sensitive to hyperparameter tuning and requires fine adjustments based on specific datasets. AdaptIDS (Elsayed et al., 2022) proposed an adaptive intrusion detection system based on LSTM and ensemble learning for aerospace-critical vehicles, which effectively improves detection of unknown attacks in complex communication environments. SNDAE (Shone et al., 2018) introduced an intrusion detection system combining Asymmetric Deep Autoencoders (NDAE) and Random Forests (RF), enhancing detection performance through automatic feature extraction. GUIDE (Yoo et al., 2024) implements a lightweight unknown attack detection system based on five GAN networks. However, the large number of networks requires significant computational resources, and the system lacks interpretability. SKT-IDS (Zha et al., 2024) builds an encoder-decoder model specifically for 0-day attack detection using Sigmoid kernel transformation, enabling training solely on known traffic and employing a cosine similarity loss function for training to detect unknown attacks. However, its complex architecture limits its interpretability. ML-based IDS (Lin et al., 2025) proposed three defense methods against

adversarial attacks on Control Area Networks: adversarial training, ensemble learning, and distance-based optimization. It also applies simulated annealing to optimize the feature space, improving classification accuracy and enabling defense against unknown adversarial attacks. However, the machine learning models used in this method are highly complex, making their decision-making process difficult to interpret, and the use of simulated annealing for feature space optimization introduces significant complexity.

However, the deep learning-based methods mentioned above generally suffer from a lack of interpretability, making their internal reasoning mechanisms difficult to explain. This reduces transparency and trustworthiness when deployed in real-world applications.

We have made a detailed comparison between IR-IDS and existing methods, as shown in Table 1. In the table, "Interpretability" refers to the interpretability of the method, i.e., the transparency of the model. "Robustness" primarily reflects the method's ability to detect novel variants of known attacks and unknow attacks. Since we simulated attacks that were not present in the training set during the experiments, we refer to these types of attacks as "unknown attacks" in this paper. It is evident that most current methods struggle to balance both interpretability and robustness effectively. Therefore, it is crucial to develop intrusion detection systems that offer both interpretability and robustness. Such systems must provide clear and transparent decision-making processes, promoting user understanding and trust in the detection results, while also demonstrating strong generalization capabilities. This ensures they can effectively identify unknown attacks in dynamic environments, thereby improving the stability and adaptability of network security defenses. To address these challenges, this paper proposes IR-IDS, an intrusion detection method that integrates both interpretability and robustness, and is capable of handling scenarios involving unknown attacks.

3. Methodology

3.1. Overview

This section provides a detailed description of the implementation process of the IR-IDS method. IR-IDS consists of two main stages: causal feature selection and robust anomaly classification. The specific framework is illustrated in Fig. 1.

The causal feature selection phase is primarily implemented using a Markov blanket search algorithm. This algorithm conducts conditional independence tests based on the causal effects of features on the target variable, evaluating the impact of each feature. By combining the results of these tests with the principles of Markov blankets, the system can select the optimal feature subset for predicting the target variable. This process ensures both the efficiency and accuracy of feature selection while introducing causal relationship analysis, allowing IR-IDS to identify and retain key features that have a direct causal influence on the target variable. As a result, noise features are effectively minimized. This causal-based feature selection approach enables the model to focus on truly meaningful features, significantly improving the accuracy and robustness of intrusion detection.

The robust anomaly classification phase mainly consists of two core components: Shapley Tree and KAN—CVAE. Shapley Tree is responsible for the pre-classification of traffic, while KAN—CVAE further distinguishes unknown anomalies. Based on the selected features, Shapley Tree optimizes the decision tree model using Shapley values. Shapley values provide a fair and interpretable method for evaluating feature contributions, quantifying each feature's impact on classification decisions. Through this optimization, Shapley Tree achieves efficient pre-classification of network traffic, significantly enhancing classification accuracy. Additionally, by assigning clear weights to each feature, Shapley Tree not only enhances detection performance but also provides highly interpretable classification results, ensuring the model's transparency and traceability.

After completing feature selection and pre-classification with Shapley Tree, IR-IDS passes the data to the combined model of the Kolmogorov-Arnold Network (KAN) and Conditional Variational Autoencoder (CVAE), which further reclassifies unknown attacks. The KAN—CVAE model detects anomalous traffic related to unknown attacks by calculating reconstruction errors. The inclusion of KAN significantly enhances the interpretability of the CVAE model, allowing it to generate clearer and more understandable latent representations, while also improving the model's robustness to unknown attacks. To further increase the reliability of anomaly detection, IR-IDS uses extreme value theory (EVT) to statistically analyze the reconstruction errors, ensuring that the detection results have high confidence. This multi-layered anomaly detection mechanism enables IR-IDS to effectively address unknown threats in complex network environments while maintaining high interpretability and robustness.

Next, we will use a case based on the content of Fig. 1 to explain the process of IR-IDS. In the IR-IDS method, the feature dataset [X_1, X_2, … X_n] and the target variable Y are first input into the system. The Markov blanket search algorithm is then applied to identify the feature subset that is causally related to the target variable Y (for example, the subset [X_4, X_5, X_6, X_8, X_9] shown in the figure). Next, the Shapley values for these feature variables with respect to Y are computed based on the decision tree model, which is then optimized to construct the Shapley Tree. In the figure, y_i represents the values corresponding to different categories of Y.

During the model training phase, a feature subset from the training set and its corresponding target variable are used as inputs to pre-train the KAN—CVAE model. In the testing phase, the test dataset is first pre-classified based on the structure of the Shapley Tree. The classification results and their corresponding feature vectors are then input into the KAN—CVAE model, where the reconstruction error of the newly generated variable X' is calculated. Finally, based on Extreme Value Theory (EVT), the reconstruction error is evaluated, and the optimized prediction result y' is output.

Through causal feature selection and the Shapley Tree, IR-IDS not only achieves fine-grained classification of known attacks but also provides transparent and intuitive decision-making logic, clearly outlining the decision process. Additionally, the KAN—CVAE model, improved with the KAN network, enhances the ability of IR-IDS to detect unknown attack anomalies while further improving the model's interpretability.

Next, we will provide a detailed explanation of IR-IDS from the perspectives of causal feature selection and robust classification algorithm.

Table 1
The capability comparison between IR-IDS and existing advanced methods.

Method	DAE	KNN	CSA-RF	SCAESVM	SVM	AGC	CGAN
Interpretability	√	√	√	√	√	√	×
Robustness	×	×	×	×	×	×	√
Method	CNN3D	AdaptIDS	SNDAE	GUIDE	SKT-IDS	ML-based IDS	IR-IDS
Interpretability	×	×	×	×	×	×	√
Robustness	√	√	√	√	√	√	√

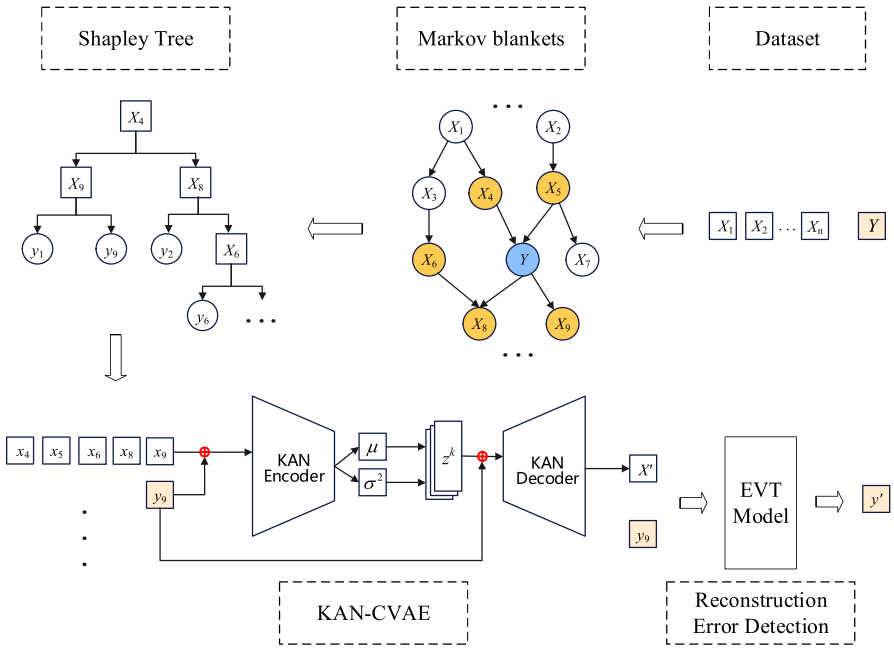


Fig. 1. Overall architecture of IR-IDS.

3.2. Causal feature selection

In IR-IDS, causal feature selection serves as the first stage of intrusion detection, significantly enhancing the efficiency and accuracy of subsequent detection processes by efficiently filtering a large number of features.

The key challenge in feature selection lies in identifying an optimal subset of features from a high-dimensional feature space that, while reducing model complexity, maintains predictive power for the target variable. Traditional feature selection methods are typically limited to analyzing the correlation or statistical dependence between features and the target variable, often overlooking deeper causal relationships between features (Wu et al., 2022). This limitation becomes especially apparent when handling complex network data, often resulting in the retention of redundant features that negatively impact model performance (Ling et al., 2023). Redundant features not only increase computational overhead but may also introduce noise, reducing the accuracy and generalization ability of classification algorithms.

To address this issue, this paper proposes an innovative causal feature selection method—DMB-PS (Dual Markov Blanket with Propensity Score). This method accurately identifies causal relationships between the target variable and feature variables through conditional independence tests based on causal effects assessment. Specifically, DMB-PS identifies the Markov blanket of the target variable to extract the smallest and most sufficient subset of features, thereby achieving optimal prediction of the target variable. The strength of this method lies in its strong adaptability and scalability, enabling seamless integration with any classification algorithm while effectively eliminating the negative impact of redundant features on model performance.

In IR-IDS, causal feature selection ensures that the selected features retain sufficient information during dimensionality reduction by modeling the causal relationships between features, thus preventing redundant features from interfering with model training. By applying the DMB-PS method, IR-IDS can efficiently select the most discriminative feature subset, significantly improving the classification accuracy of the intrusion detection system and laying a solid foundation for anomaly detection of unknown attacks.

Experimental results show that, through precise causal feature selection, IR-IDS can significantly streamline the feature set, optimizing the performance of subsequent classification algorithms. This not only improves the system's efficiency in recognizing known attack types but also enhances its ability to detect unknown attacks. The introduction of the DMB-PS method enables IR-IDS to demonstrate greater robustness in complex network data environments, providing a more reliable technical foundation for intrusion detection systems.

The following sections will first introduce the concept of the Double Markov Blanket and then explain how conditional testing is performed based on the theory of propensity score.

3.2.1. Double markov blanket

The Markov Blanket search algorithm is a causal inference-based feature selection method designed to identify the Markov Blanket of a target variable (Wu et al., 2022). By isolating a subset of features that are causally dependent on the target variable, it ensures that all other variables become conditionally independent of the target variable given this subset. Because these algorithms leverage causal information, they are often referred to as causal feature selection methods.

However, traditional Markov Blanket search algorithms often emphasize precision while neglecting recall, which contradicts the principle of "better too many than too few" in intrusion detection scenarios (Ling et al., 2023). To address this limitation, this paper proposes an efficient new method for identifying the Markov Blanket, which not only accelerates the search process but also ensures the completeness of the selected features. The following definitions provide the necessary background.

Definition 1. Bayesian Network (Andersen, 1991). A Bayesian Network is defined as a triplet (U, G, P), where U denotes a set of variables, G represents a Directed Acyclic Graph (DAG) over U, P is a probability distribution over U. For any variable X in U, if the parents of X in G are used as the conditioning set, and all non-descendants of X are conditionally independent of X in P, then (U, G, P) is a Bayesian Network.

Definition 2. Faithfulness (Andersen, 1991). Given a Bayesian Network (U, G, P), the graph G is faithful to the distribution P if and only if every conditional independence relationship in P is determined by G and the Markov condition. Conversely, P is faithful to G if and only if there exists a subgraph of G that is faithful to P.

Assuming that a Bayesian Network satisfies the faithfulness

condition, any independence relationships within the network are not due to coincidental patterns but arise from the network's structural properties. In the context of intrusion detection, a Bayesian Network can be constructed to model the causal relationships among all features based on observable data. Since all observable data are generated through deterministic processes, the causal relationships among features can be considered unique. In other words, the structure of the Bayesian Network can be uniquely determined from observational data, accurately capturing the conditional dependencies among features. Therefore, the intrusion detection problem satisfies to the faithfulness assumption.

Definition 3. **Markov Blanket** (Andersen, 1991). A Markov blanket $MB_I(X)$ of an element $X \in U$ and $X \notin S$ is any subset S of elements for which

$$I(X, U - S - X|S) \tag{1}$$

A set is called a Markov boundary of X, denoted $B_I(X)$, if it is a minimal Markov blanket of T, i.e., none of its proper subsets satisfy Eq. (1).

Here, $I(X,\ U - S - X|S)$ is defined as representing the conditional independence between variable X and $U - S - X$ given S, based on their probability distribution.

Definition 4. **Optimal Solution for Feature Selection.** In machine learning and statistical modeling, the optimal solution for feature selection refers to selecting a subset of features that best represents the target variable.

Theorem 1. (Pellet and Elisseeff, 2008): *In data that satisfies the faithfulness assumption, the Markov Boundary of the target variable is unique and serves as the optimal solution for feature selection.*

The proof of Theorem 1 consists of two main parts:

a) Uniqueness of the Markov Boundary. Under the faithfulness assumption, the data is assumed to originate from a distribution that conforms to a specific graph structure. According to this assumption, the edges in the graph represent definitive conditional dependency relationships between variables. Since the conditional independence relationships of each variable are strictly encoded by the graph structure, the Markov Boundary of the target variable can be uniquely determined by the nodes that are directly connected to the target variable in the graph. Now, suppose there exist two distinct feature subsets, S_1 and S_2, both of which satisfy the conditions of the Markov Boundary, meaning they both form conditional independence sets for the target variable. For this to hold, both subsets must include all features that are directly related to the target variable. However, since the graph structure is unique and the conditional independence relationships between features are uniquely defined, it is impossible for two different subsets to simultaneously satisfy the conditions of the Markov Boundary. Therefore, the Markov Boundary of the target variable is guaranteed to be unique.

b) The Markov Boundary as the Optimal Solution for Feature Selection. In feature selection, the objective is to identify the optimal subset of features that allows for the most accurate prediction of the target variable. Under the faithfulness assumption, the Markov Boundary of the target variable is the smallest set of features that includes all those conditionally dependent on the target variable. Any feature not included in the Markov Boundary is conditionally independent of the target variable when the Markov Boundary is given. Therefore, the Markov Boundary is the optimal solution for feature selection. It represents the minimal feature subset that contains all the necessary information required to ensure the best possible prediction of the target variable, given these features.

Theorem 1 provides a theoretical guarantee for using Markov Boundary discovery algorithms to solve feature selection problems. According to Definition 1 and Definition 3, in a Bayesian network that satisfies the faithfulness assumption, the Markov Boundary of a node consists of its parent nodes, child nodes, and spouse nodes (i.e., other parents of its child nodes) (Andersen, 1991). The Markov Blanket, however, is a superset of the Markov Boundary.

As illustrated in Fig. 2, part (a) shows a Bayesian network with 10 nodes, where node 7 is the target node. In part (b), the highlighted yellow nodes represent the Markov Boundary of node 7, which includes its parent nodes {4, 5}, child nodes {9, 10}, and spouse node {6}. In part (c), additional nodes {3, 8} are included, forming the highlighted yellow nodes {3, 4, 5, 6, 8, 9, 10}, which represents the Markov Blanket of node 7. In essence, the Markov Blanket is formed by augmenting the Markov Boundary with any additional nodes.

The Markov Boundary is often challenging to rigorously prove in theoretical terms. Consequently, most research has focused on developing various Markov Blanket search algorithms to approximate solutions that satisfy the required conditions. These algorithms can be broadly categorized into constraint-based methods, score-based methods, and structure-based methods (Wu et al., 2022). Among them, constraint-based Markov Blanket search algorithms typically utilize statistical methods to validate causal relationships between variables. Compared to the other two approaches, constraint-based methods offer greater interpretability and a more robust theoretical foundation, making them particularly well-suited for high-dimensional scenarios and widely adopted in practice.

Traditional Markov Blanket search algorithms focus primarily on the precision and accuracy, often requiring substantial computational time and resulting in a minimal set of features. However, in IDS, achieving high recall and computational efficiency is more critical to ensure classification accuracy and effective search performance. Therefore, our objective is to identify the Markov Blanket of the target variable with the highest possible recall while maintaining efficiency and practicality. The final goal of the algorithm is to ensure that, for $S = U - Y$, any $X \in S \backslash \mathrm{MB_Y}$:

$$X \perp Y | MB_Y \tag{2}$$

To address this challenge, we propose a more efficient Markov Blanket discovery algorithm called Double Markov Blanket (DMB). This algorithm operates in two phases: Growth and Shrink (Dou et al., 2015).

In the Growth phase, the algorithm searches through all feature variables based on the current candidate Markov Blanket CMB_G, adding features that are not independent of the target variable to the conditional feature set.

In the Shrink phase, the algorithm iteratively examines the conditional feature set collected in the previous phase to refine the candidate Markov Blanket CMB_S. During each iteration, the conditional feature set is used as a condition to remove features that are independent of the target variable under the given conditions. The algorithm terminates when no further changes occur in the conditional feature set.

The main steps of the method are detailed in Algorithm 1, where $X \perp Y$ denotes that X is independent of Y.

3.2.2. Condition testing method based on propensity score

From the workflow of the Double MB algorithm, it is clear that the accuracy and efficiency of conditional testing are crucial to the algorithm's overall performance. Traditional conditional testing methods, such as the G^2 test and mutual information, face significant challenges in high-dimensional spaces, where data distributions become exceedingly sparse. This leads to the well-known "curse of dimensionality", resulting in a substantial decline in accuracy and compromising the reliability of experimental results. To address this issue, we propose a novel conditional testing method based on propensity score. To the best of our knowledge, this is the first study to apply propensity score in the context of intrusion detection.

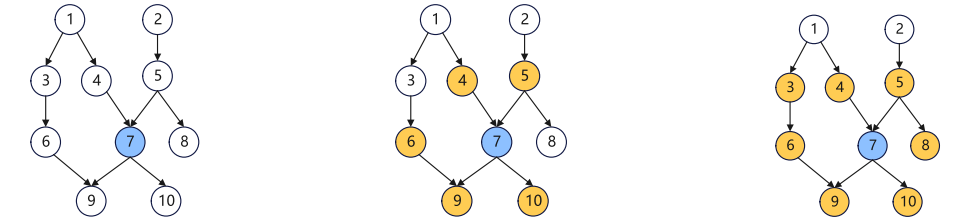


Fig. 2. Markov Boundary and Markov Blanket.

Algorithm 1
Double Markov Blanket.

Input: All feature variables S, target feature variable Y
Output: Markov Blanket of Y (MB_Y)
1: CMB_G= [], Flag←1
2: for X in S:
3: if $X \perp Y \mid CMB_G$:
4: S.remove(X)
5: else:
6: CMB_G.append(X)
7: $CMB_S \leftarrow CMB_G$
8: While Flag:
9: $K \leftarrow$ len(CMB_S)
10: for X in CMB_S:
11: if $X \perp Y \mid CMB_S$:
12: CMB_S.remove(X)
13: if K = len(CMB_S): Flag←0
14: $MB_Y \leftarrow CMB_S$

The propensity score is a statistical method widely used in causal inference to address potential confounding bias in observational studies. Its primary objective is to estimate the likelihood of an individual receiving a particular treatment (or intervention) based on observed covariates, thereby mitigating bias caused by confounding variables. This enables a more accurate evaluation of the causal effect of the treatment on the outcome. The definition is as follows:

Definition 5. **Propensity Score** (Rosenbaum and Rubin 1983). For an individual i, the propensity score $e(T_i)$ represents the probability of being assigned to the treatment group, conditional on the observed covariates X_i, and is expressed as:

$$e(T_i) = P(X_i = 1 | T_i) \tag{3}$$

Here, X_i represents whether an individual receives the treatment, where $X_i = 1$ indicates the individual receives the treatment, and $X_i = 0$ indicates the individual does not receive the treatment. The index i refers to sample i. For sample i, the effect of the treatment on the outcome is defined as:

$$\tau_i = Y_i(1) - Y_i(0) \tag{4}$$

Here, Y_i1 represents the potential outcome for sample i if it receives the treatment, and Y_i0 represents the potential outcome if it does not receive the treatment. These are referred to as potential outcomes because, for any given sample i, only one outcome can be observed: either receiving the treatment or not receiving the treatment.

Thus, by representing the treatment state for a single variable as $T = 1$ and the non-treatment state as $T = 0$, the Average Treatment Effect (ATE) is defined as the expected treatment effect across all samples. It is calculated as follows:

$$ATE = E[Y_i(T = 1) - Y_i(T = 0)] \tag{5}$$

In propensity score analysis, the ATE can be used as a metric to evaluate the causal effect of a specific treatment on the outcome. Based on this idea, this paper proposes a conditional testing method based on causal effects for feature selection in intrusion detection.

As shown in Eqs. (3)-(5), traditional propensity score calculations consider the dependent variable to have only two states: treatment received or treatment not received. This approach is not suitable for multi-class detection problems. Additionally, traditional propensity score models typically use matching methods to align potential outcomes, thereby reducing the influence of covariates. However, in intrusion detection, it is unnecessary to modify the existing observable data. Instead, the goal is to determine the effect of changes in independent variables on dependent variables under the current covariate conditions. This is because, under the same covariate conditions, if the independent variable does not exhibit significant changes, it indicates that the information carried by the independent variable can be fully replaced by the covariates. Based on these considerations, this paper introduces improvements to the traditional method.

First, data preprocessing is required. Since this is a multi-class problem, all feature variables need to be classified according to the total number of categories, $|NC|$. This paper uses the K-means clustering method. To better highlight anomalies in the features, clustering is performed based on the degree of abnormal deviation in values. Specifically, the data in the j-th column C_j is first divided into two clusters ($K = 2$), and the optimization objective function is defined as follows:

$$\min \sum_{k=1}^{K} \sum_{c \in C_j} \| c - \mu_k \|^2 \tag{6}$$

In this case, $K = 2$, meaning there are only two categories: normal and abnormal., μ_k represents the centroid of the k-th category. Then, the category whose centroid is closer to the overall mean is selected as the normal category:

$$Normal_Cluster = \arg\min_k |\mu_k - \overline{C}_j| \tag{7}$$

Subsequently, the data points within the normal category are assigned a label of 0. The abnormal category is then identified based on Eq. (6), and the anomaly score for data point c_i is calculated as:

$$SC_i = c_i - \mu_0 \tag{8}$$

Finally, the anomaly points are ranked according to their degree of abnormality:

$$L_i^{level} = \left\lceil \frac{SC_i}{\max(SC)} \times |NC| \right\rceil \tag{9}$$

After classifying the data into different levels, the conditional independence of each feature X is tested under the Candidate Markov Blanket (CMB) set of the target variable Y. Since the numerical values of Y only represent different categories without conveying specific meanings, making judgments based on the magnitude of Y is inaccurate. Instead, this study uses the frequency of different values of Y as the basis for evaluation. By determining the variable values in the CMB and comparing the frequency distributions of Y across different values of X, the causal effect of X on Y can be more accurately reflected.

To begin, the values of CMB need to be determined. Let the total number of feature columns be N. Then, the number of columns in CMB is

$n < N$, and its values are defined as:

$$CMB = [C_1, C_2, \ldots C_n] \rightarrow \left[c_{k_1}^1, c_{k_2}^2, \ldots c_{k_n}^n\right] \tag{10}$$

Where $0 \le c_{k_i}^i \le |NC| - 1$. Once CMB is determined, the frequency of Y when X takes the value x_i is evaluated as:

$$P(y_i, x_i) = P\left[Y = y_i, X = x_i \middle| CMB_i = \left(c_i^1, c_i^2, \ldots c_i^n\right)\right] \tag{11}$$

Thus, the current distribution sequence is:

$$PY_{x_i} \middle| CMB_i = \left[P\left(y_0, x_i\right), P(y_1, x_i), \ldots P\left(y_{NC}, x_i\right)\right] \middle| (X = x_i | CMB_i) \tag{12}$$

The number of PY_{x_i} depends on the number of distinct values of X under a specific CMB_i. After calculating all PY_{x_i}, the Kullback-Leibler (KL) divergence is computed pairwise, and the maximum value is taken as the measure of the causal effect of X on Y under CMB_i. The calculation formula is as follows:

$$D_{KL}^{CMB_i} = \max D_{KL}\left(PY_{x_i} \parallel PY_{x_j}\right) = \max \sum PY_{x_i} \log \frac{PY_{x_i}}{PY_{x_j}} \tag{13}$$

Finally, the maximum value of all $D_{KL}^{CMB_i}$ is taken as the overall causal effect of X on Y under the CMB:

$$CE(X, Y|CMB) = \max_{CMB} D_{KL}^{CMB_i} \tag{14}$$

The meaning of $CE(X, Y|CMB)$ is the maximum difference in the distribution of Y caused by the different values of X under the determined condition set CMB. This measure effectively reflects the causal effect of X on the overall distribution of Y under the given condition set and is not influenced by the specific values of Y. Therefore, when CE is smaller than a certain threshold, it is considered that X does not have a significant causal effect on Y, and the two are conditionally independent under CMB. The threshold is determined using percentiles, as follows:

$$CE < percentile(CE(X, Y|CMB), \alpha) \Rightarrow X \perp Y | CMB \tag{15}$$

The conditional testing method based on propensity score effectively addresses the curse of dimensionality, ensuring accurate computational results while avoiding the influence of Y 's value settings. Compared to traditional conditional testing methods, it is better suited for multi-class scenarios. Moreover, it offers stronger interpretability compared to other machine learning-based feature extraction methods.

3.3. Robust anomaly classification

Based on causal feature selection, the second phase of IR-IDS achieves robust classification. This phase consists of two main parts: pre-classification and unknown attack reclassification. The pre-classification part uses a Shapley value-based decision tree model to perform fine-grained classification of network traffic, while the unknown attack division part combines the Kolmogorov-Arnold Network (KAN) and Conditional Variational Autoencoder (CVAE), detecting anomalous traffic through reconstruction error analysis and effectively reclassifying unknown attacks.

We name the Shapley value-optimized decision tree model as "Shapley Tree." In this part, the Shapley value quantifies the contribution of each feature to the decision tree's prediction and optimizes the model based on this. The optimized Shapley Tree not only retains the inherent interpretability of the decision tree but also significantly improves its prediction performance. The introduction of Shapley values clearly shows the impact of each feature on the decision-making process, resulting in a transparent classification mechanism. This method provides traceable explanations for each prediction, enhancing the model's credibility and understandability in real-world applications.

Furthermore, by combining the Kolmogorov-Arnold Network (KAN) with the traditional Conditional Variational Autoencoder (CVAE), the model's ability to express and perform in high-dimensional, complex data is significantly improved. CVAE itself has strong reconstruction capability and can capture the latent distribution of input data, enabling effective anomaly detection. The inclusion of KAN further enhances the model's ability to process and generate latent data representations, providing more precise support for the detection of unknown attacks. The introduction of KAN not only improves the model's robustness but also enhances its interpretability, avoiding the common limitations of black-box models, ensuring that each detection result can be clearly analyzed and understood.

By combining these two components, this method not only avoids the limitations of black-box models but also achieves fine-grained classification of known attacks and effective anomaly detection for unknown attacks.

The following will provide a detailed description of the robust classification algorithm, focusing on the Shapley Tree and KAN—CVAE components.

3.3.1. Shapley tree

The Shapley value (Yin et al., 2022) is a distribution method originating from cooperative game theory, with the core objective of fairly distributing the collective gains created by multiple participants by quantifying each individual's contribution to the team. The essence of the Shapley value is to measure the importance of each feature in model decision-making based on its marginal contribution to the prediction result.

Because the Shapley value can precisely quantify the impact of each feature on model predictions, it has gained widespread application in the field of model interpretability in machine learning in recent years. The key criteria for evaluating machine learning interpretability models are primarily based on three core properties (Lundberg and Lee 2017): *local accuracy, consistency*, and *missingness*.

Local accuracy ensures that the sum of the Shapley values for all features equals the model's prediction for a specific sample. This indicates that the prediction can be fully decomposed into the sum of the contribution values of individual features, ensuring the accuracy of feature attribution. Local accuracy is one of the key distinctions of the Shapley value compared to other feature attribution methods. It guarantees a fair allocation, making the Shapley value a theoretically rigorous and highly interpretable attribution method. The formula is as follows:

$$f(x) = \phi_0(f) + \sum_{i=1}^{|M|} \phi_i(f, x) \tag{16}$$

Here, x represents a specific sample, $f(x)$ denotes the model's prediction output, $\phi_i(f, x)$ is the Shapley value of feature i, $\phi_0(f)$ is the baseline value, which corresponds to the average output of the baseline model on the background data, M is the set of all features, and $|M|$ is the total number of features.

Consistency indicates that if the contribution of feature i increases in the model, its Shapley value should also increase. That is, for any two models f and f', if:

$$f'_x(S) - f'_x(S \backslash i) \ge f_x(S) - f_x(S \backslash i) \tag{17}$$

Then:

$$\phi_i(f', x) \ge \phi_i(f, x) \tag{18}$$

Here, $S \subseteq \mathscr{F}$, and $\mathscr{F}$ represents the power set of M. Consistency ensures that if a model is modified such that the contribution of certain features increases or remains unchanged, the attribution for these features should not decrease, regardless of other inputs. This guarantees the monotonicity of the Shapley value.

Missingness indicates that if the inclusion of feature i does not affect the model's output, then its Shapley value should be 0 for any feature combination. Specifically, if:

$$f_x(S \cup i) = f_x(S) \tag{19}$$

Then:

$$\phi_i(f,x) = 0 \tag{20}$$

The classic Shapley value satisfies all these properties simultaneously. Its calculation formula is as follows:

$$\phi_i(f,x) = \sum_{R\subseteq\Re} \frac{1}{|M|!}\left[f_x\left(P_i^R \cup i\right) - f_x\left(P_i^R\right)\right] \tag{21}$$

Here, $\mathscr{R}$ represents all possible orderings of features. Note that $\mathscr{R}$ is different from the power set $\mathscr{T}$, as the sets in $\mathscr{R}$ depend on the sequence of features in M. Thus, $|\mathscr{R}| = |M|!$, whereas $|\mathscr{T}| = 2^{|M|}$. P_i^R represents the subset of all features preceding feature i in the ordering $\mathscr{R}$. $f_x(S)$ denotes the predicted value based on the feature values of subset S for sample x. The term $f_x\left(P_i^R \cup i\right) - f_x\left(P_i^R\right)$ represents the marginal contribution of feature i to the prediction in the ordering $\mathscr{R}$.

As shown in Eq. (21), calculating the classic Shapley value requires traversing all possible feature subsets, resulting in an exponential increase in computational complexity as the size of the set grows. To address this issue, TreeExplainer (Lundberg et al., 2020) introduced the first polynomial-time path-dependent feature perturbation algorithm tailored for tree structures. This algorithm calculates path weights during each recursion to determine the contribution of each feature subset to the prediction.

This method allows TreeExplainer to comprehensively and accurately quantify the importance of each feature, providing a robust foundation for model transparency and reliability. Based on this, the study combines TreeExplainer's results with decision trees, introducing a Shapley-based decision algorithm called Shapley Tree. This algorithm reorganizes the feature positions in the decision tree according to their Shapley values, resulting in an interpretable classification model that is further extended to multi-class scenarios for fine-grained intrusion detection. Compared to traditional methods that use information gain and entropy, this approach is more aligned with practical scenarios, as the importance of features is more directly related to the target variable, thereby yielding more precise results.

TreeExplainer optimizes the Shapley value calculation algorithm for tree structures, reducing computational complexity to polynomial time ($O\left(TLD^2\right)$), where T represents the number of trees, L the number of leaf nodes, and D the depth of the tree. The core idea involves progressively decomposing path contributions from the root to the leaf nodes, with the key being the dynamic update of path weights. At each split node, the path weights are extended to the hot branch and cold branch based on the sample's splitting condition. Once a leaf node is reached, the prediction value of the leaf is allocated to the contributions of features along the path according to the calculated weights, ultimately deriving the Shapley values. During the recursion's return phase, the extension operations are reversed to restore the state, enabling efficient decomposition and accumulation of feature contributions within polynomial time. The calculation formula can be summarized as follows:

$$\phi_e = \sum_{\left(p,e_{leaf}\right)} m[p] \times f\left(e_{leaf}\right) \tag{22}$$

Here, $p \in \text{path}\left(e \to e_{leaf}\right)$ represents the path from node e to its corresponding leaf node $f\left(e_{leaf}\right)$ is the prediction value of the leaf node, and $\text{m}[p]$ represents the weight of the path. The weight of the path is calculated based on the split proportion. If the current path weight is m_i, then the weight of its branch k is updated as:

$$m_{k,i} = m_i \times P_k \tag{23}$$

Here, P_k represents the split ratio of branch k, and its calculation formula is as follows:

$$P_k = \frac{N_k}{N_e} \tag{24}$$

Here, N_k represents the number of samples in branch k, and N_e is the total number of samples at the current node.

Using Eqs. (16) to (24), the Shapley value corresponding to all features in the tree model can be computed. By weighting the Shapley values of all prediction values, the total Shapley value for feature i is obtained as follows:

$$\Phi_i = E\left(\phi_i^j\right) \tag{25}$$

Here, ϕ_i^j represents the Shapley value of feature i for class j. Algorithm 2 provides an overview of the Shapley Tree's overall process. The input to the algorithm is the training dataset, and the output is the Shapley Tree model.

First, the model is initialized as a decision tree. Although Shapley values can quantify the importance of features, they cannot provide decision-making criteria, so a decision tree model is still required for making decisions. After initialization, the Shapley values of features for each class are calculated, where NC represents all classification categories. The RECURSE function serves as the core process for calculating Shapley values based on the decision tree. This function traverses all child nodes of the current node. If a child node is a leaf node, the current weight and prediction value are returned to the parent node; otherwise, the RECURSE function is called recursively to traverse the child nodes.

Finally, the Shapley value expectations for each feature are calculated based on the Shapley values across all classes. Features are ranked according to their expectations, and the model is retrained to generate the Shapley Tree model.

3.3.2. Kolmogorov–Arnold networks and conditional variational autoencoder

The Shapley Tree enables fine-grained classification of traffic but does not take unknown attacks into account. To address this limitation, this study further refines the prediction results by identifying and re-evaluating unknown attacks.

It is important to emphasize that the ultimate goal of this study is to achieve fine-grained classification of known traffic and anomaly detection of unknown attacks, without distinguishing the specific categories of the unknown attacks. To achieve this, the reconstruction error of Conditional Variational Autoencoder (CVAE) (Sohn et al., 2015) is combined with the Extreme Value Theory (EVT) method to detect unknown attacks. Furthermore, to enhance the interpretability of the model, the Kolmogorov–Arnold Networks (KAN) (Liu et al., 2025) is used as a replacement for traditional neural network models, improving the performance of the CVAE.

Kolmogorov–Arnold Networks (KAN) is a neural network model based on mathematical theory, specializing in modeling complex nonlinear data. By mapping multidimensional inputs into a simplified form, it efficiently predicts complex time series, particularly useful for handling nonlinear trends. Compared to traditional multilayer perceptron (MLP), KAN offers greater interpretability, grounded in Kolmogorov's super approximation theorem. This theorem states that any multidimensional continuous function can be decomposed into a sum of one-dimensional continuous functions. This feature allows KAN to represent complex multidimensional nonlinear functions with fewer one-dimensional functions, simplifying the model's structure and making the prediction process more transparent.

This study develops a CVAE model based on the KAN network, using reconstruction error to evaluate the differences between predicted traffic and training traffic, thereby identifying misclassified unknown attack traffic. The training objective of CVAE is to maximize the Evidence Lower Bound (ELBO), which enables data modeling through variational inference. The main structure of the KAN—CVAE network is shown in Fig. 3:

In the figure, x represents the feature vector, and y represents the label data. After concatenating x and y, they are used as the input to the KAN Encoder, which extracts the corresponding mean μ and variance σ^2.

Algorithm 2
Shapley Tree.

Input: Train Dataset (X, y)
Output: Model Shapley_Tree
1: init: model = DecisionTree(X), $m = 1$, $\phi_i^j = 0$(i ∈ M,j ∈ NC)
2: for j in NC:
3: procedure RECURSE(e):
4: $m_{k,i} = m_i \times P_k$
5: for k in children(e):
6: if k is leaf: $\phi_e^j = \phi_e^j + m_{k,i} \times fk$
7: else:
8: RECURSE(k)
9: $\Phi_i = E\left(\phi_i^j\right)$
10: Shapley_Tree = Sort(X, Φ_i) in model

These parameters define a latent distribution:

$$q_\Phi(z|x,y) = \mathbb{N}\left(z;\mu,\sigma^2\right) \tag{26}$$

Subsequently, the reparameterization trick is used to sample Z^k from the latent distribution:

$$z^k = \mu + \sigma \cdot \varepsilon \tag{27}$$

Here, ε represents the sampling noise drawn from a standard normal distribution. This technique decouples the sampling process of z^k from the network parameters, enabling gradient propagation. Finally, the latent variable z^k is combined with the conditional label y and input into the KAN Decoder for decoding. The reconstructed data x^k corresponding to the label y is then generated. By comparing $\hat{x}^k$ with x, the loss function is calculated based on the log-likelihood, and the model is trained accordingly:

$$L(\theta,\Phi;x,y) = E_{q_\Phi(z|x,y)}[\log p_\theta(x|z,y)] - D_{KL}(q_\Phi(z|x,y) \parallel p(z|y)) \tag{28}$$

To detect unknown attacks, the network is first trained on the training dataset to ensure it captures the feature distributions of known classes. For the test dataset, the trained network is used to perform reconstruction, and the reconstruction error is computed.

In this study, KL divergence is used as the metric for reconstruction error, as it quantifies the distributional difference between the input data and the reconstructed data. For unknown data, the network struggles to accurately recover the features during reconstruction, leading to significantly higher reconstruction errors.

To identify such anomalous data, this study leverages Extreme Value Theory to analyze the distributional properties of reconstruction errors. Extreme Value Theory is a statistical framework often applied in computer vision to model the probability of extreme events, such as the tail distribution of errors. In this approach, the tail distribution of reconstruction errors is modeled using the Generalized Pareto Distribution (GPD), which effectively characterizes errors exceeding a specific threshold. This method enables dynamic adjustment of the threshold, making it well-suited to handle complex data distributions in open-world scenarios.

4. Experiments and analysis

To validate the effectiveness of the IR-IDS, we implemented the system using Python. The experimental setup featured an Intel(R) Xeon (R) W-2225 processor with a clock speed of 4.1 GHz and 32 GB of memory. In the feature selection phase, the independence threshold was set to 0.75.

For the classification algorithm's parameter settings, we adopted a systematic approach to determine the key hyperparameters. First, the initial distribution threshold was set to 0.9, a choice based on preliminary experimental validation, which effectively balanced the model's sensitivity and specificity. For the training duration of the KAN—CVAE model, we performed meticulous tuning using five-fold cross-validation. During this process, we tested training durations ranging from 10 to 50 epochs and observed that when the number of epochs reached 20, the validation loss stabilized, and performance metrics (including accuracy and F1 score) reached a steady state. As a result, we set the final training epoch to 20.

Regarding the design of the KAN architecture, we conducted a comprehensive structural search experiment. Specifically, we evaluated different depth configurations, ranging from 1 to 3 hidden layers, as well as various width combinations with 8 to 64 nodes. Through grid search and performance comparison analysis, we ultimately determined that a single hidden layer with 12 nodes was the optimal configuration. This choice was based on the following considerations: (1) it achieved stable classification accuracy on the validation set; (2) compared to deeper or wider architectures, this configuration exhibited lower generalization error on the test set; (3) the smaller network size effectively prevented overfitting; and (4) the streamlined network structure significantly improved the model's interpretability, making feature importance analysis more intuitive and reliable. This optimized architecture not only ensured model performance but also met the practical requirement for model interpretability in real-world applications.

A total of four sets of experiments were conducted: (4.2) a comparison of IR-IDS with other intrusion detection methods, (4.3) an ablation study, (4.4) a comparison of feature selection methods, and (4.5) a comparison of classification algorithms.

4.1. Dataset

This study uses five publicly available intrusion detection datasets: CIC-IDS2017, CSE-CIC-IDS2018, CIC-DDoS2019, CIC-UNSW-NB15, and the newly released CIC-IoT-IDAD-2024 (data source: https://www.unb.ca/cic/datasets/index.html). These datasets cover a variety of network attack types, such as Denial of Service (DoS) attacks, worm attacks, and malware propagation, providing a comprehensive assessment of the proposed system's performance across different attack scenarios. Moreover, these datasets are widely recognized in the field of cybersecurity and have been extensively used in numerous intrusion detection studies, ensuring the reliability and reproducibility of the experimental results. At the same time, as the comparison algorithms in existing studies are mostly evaluated using these datasets, our selection

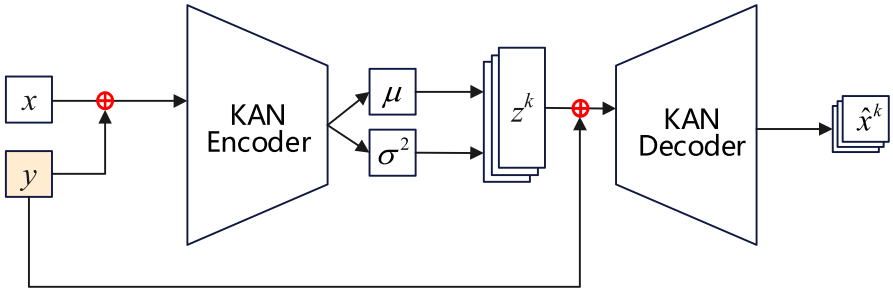


Fig. 3. Structure of the KAN—CVAE Network.

guarantees the fairness of the experiment, ensuring that performance comparisons between different methods are consistent and objective. To further validate the applicability of the proposed method in emerging scenarios, we also introduced the newly released CIC-IoT-IDAD-2024 dataset, addressing the evolving new challenges in the current cybersecurity landscape.

During the sampling process, the training set was created by sampling according to the proportion of traffic in the overall dataset, with benign traffic making up approximately half of the data.

After removing erroneous entries, the final sampling results are presented in Tables 2-4. These tables show the proportion of different attack types in each dataset, as well as the total number of records in the training sets. To assess the algorithm's capability to detect unknown attacks, each test set contains at least 10 % unknown attacks. Specifically, the IDS2017 training set includes 9 attack types, with 6 unknown attacks in the test set. The IDS2018 training set includes 9 attack types, with 6 unknown attacks in the test set. The DDoS2019 training set contains 6 attack types, while the test set includes 8 unknown attack types. The UNSW-NB15 training set contains 5 attack types, with 5 unknown attack types in the test set. The IoT-IDAD-2024 training set contains 8 attack types, with 8 unknown attack types in the test set.

The evaluation metrics used in this paper primarily focus on attack detection accuracy. Additionally, Precision, F1-score, and Recall are included in the comparison of overall methods and classification algorithms, while the number of selected features and the time taken are considered in the comparison of feature selection algorithms.

4.2. Comparison experiment

In this paper, we conducted comparative experiments for both multi-class and binary classification tasks using multiple datasets. The comparison metrics include Precision, F1, Recall, and Accuracy. The experimental results are presented in Tables 5–9, with the best result in each column highlighted in bold. We compared the proposed IR-IDS method with several existing intrusion detection algorithms, including KNN (Pajouh et al., 2019), SVM (Jeong et al., 2022), CSA-RF (Mohiuddin et al., 2023), CGAN (Wu et al., 2023), CNN3D (Yue et al., 2022), DAE (Megantara and Ahmad, 2021), AdaptIDS (Elsayed et al., 2022), SCAESVM (Wang et al., 2020), and SNDAE (Shone et al., 2018). Additionally, to assess the improvements introduced by our method, we also compared it with the baseline Decision Tree (DT) model and the KAN network (KAN) used individually.

The primary goal of this paper is to achieve fine-grained classification for known attacks and anomaly detection for unknown attacks. Therefore, when evaluating prediction accuracy, we ensure that attacks present in the training set are classified correctly into their respective categories, while attacks not seen during training are detected as anomalies.

From the experimental results in the four tables, it is clear that IR-IDS performs excellently in both multi-class and binary classification tasks, demonstrating stable performance across all four datasets. When compared with state-of-the-art methods, IR-IDS showed up to a 48.65 % improvement in multi-class accuracy, with the smallest improvement being 3.02 %, highlighting the effectiveness of our approach. In comparison to the baseline models KAN and DT, IR-IDS improved accuracy by as much as 69.41 %, with at least a 4.96 % increase, demonstrating the significant improvements made to the basic models. This gap is particularly evident in the latest dataset, IoT-IDAD-2024, which may be due to the fact that current algorithms have not been applied to the most recent datasets. However, it also indirectly reflects the robustness and generalization ability of IR-IDS.

Although the performance gap is smaller in binary classification tasks, our method still ranks highly. IR-IDS achieved the highest accuracy in most cases, with the exception of the UNSW-NB15 dataset, where it ranked second.

Furthermore, during the experiments, we observed that IR-IDS, along with some other algorithms such as SNDAE and DT, showed a slight decrease in accuracy in binary classification compared to multi-class classification. Upon further investigation, we determined that this drop in accuracy may be due to overfitting, which is more likely to occur in binary classification when the number of classes is smaller. To address this issue, implementing an early stopping mechanism could prevent overfitting by halting training when the model performance starts to degrade, leading to better classification accuracy.

4.3. Ablation experiments

To assess the contribution of different modules in IR-IDS, we conducted ablation experiments focusing solely on multi-class classification, with prediction accuracy as the evaluation metric. The results across four datasets are presented in Tables 10–14. We divided the method into three components for ablation: DMB-PS represents the feature selection module, Shapley-Tree denotes the fine-grained classification module, and KAN—CVAE corresponds to the unknown attack detection module. A "$\surd$" indicates that the module was included, while a " $\times$ " indicates that it was excluded. However, to ensure proper input for KAN—CVAE when the Shapley-Tree module is excluded, we substitute it with the base decision tree model. The evaluation metric is prediction accuracy, with "Overall Accuracy" representing the total accuracy, followed by the accuracy of various attack types and normal traffic, where bold entries indicate unknown attacks.

The overall accuracy results show that excluding certain modules

Table 2
Sampling information for the IDS2017 and IDS2018 datasets.

Index	IDS2017			IDS2018		
	Attack Category	TRAIN	TEST	Attack Category	TRAIN	TEST
1	BENIGN	49.61 %	52.10 %	Benign	48.11 %	51.80 %
2	DoS Slowhttptest	6.73 %	3.54 %	SSH-Bruteforce	7.49 %	5.43 %
3	DDoS	6.73 %	4.39 %	DoS attacks-GoldenEye	7.49 %	3.11 %
4	FTP-Patator	6.73 %	4.41 %	DoS attacks-SlowHTTPTest	7.49 %	5.43 %
5	SSH-Patator	6.73 %	8.82 %	DDOS attack-HOIC	7.49 %	5.24 %
6	DoS slowloris	6.73 %	4.41 %	Infilteration	7.49 %	3.15 %
7	PortScan	6.72 %	4.41 %	Bot	7.49 %	5.43 %
8	DoS Hulk	6.70 %	4.40 %	DDoS attacks-LOIC—HTTP	5.61 %	1.36 %
9	Web Attack Brute Force	3.31 %	2.26 %	DoS attacks-Hulk	1.35 %	4.87 %
10	Bot	0.00 %	1.49 %	DoS attacks-Slowloris	0.00 %	2.32 %
11	DoS GoldenEye	0.00 %	7.65 %	FTP-BruteForce	0.00 %	0.46 %
12	Infiltration	0.00 %	0.11 %	Brute Force -Web	0.00 %	5.43 %
13	Web Attack XSS	0.00 %	1.92 %	Brute Force -XSS	0.00 %	0.42 %
14	Web Attack Sql Injection	0.00 %	0.06 %	SQL Injection	0.00 %	0.13 %
15	Heartbleed	0.00 %	0.03 %	DDOS attack-LOIC-UDP	0.00 %	5.43 %
	Total Count	22,286	33,995	Total Count	20,032	27,631

Table 3
Sampling information for the DDoS2019 and UNSW-NB15 datasets.

Index	DDoS2019			UNSW-NB15		
	Attack Category	TRAIN	TEST	Attack Category	TRAIN	TEST
1	BENIGN	50.00 %	54.75 %	Normal	52.51 %	49.85 %
2	DrDoS_UDP	12.52 %	2.97 %	Fuzzers	12.03 %	3.09 %
3	DrDoS_LDAP	12.50 %	4.14 %	Generic	11.92 %	5.49 %
4	Portmap	12.50 %	4.14 %	DoS	11.88 %	2.75 %
5	TFTP	6.25 %	4.14 %	Reconnaissance	11.67 %	8.96 %
6	DrDoS_MSSQL	6.24 %	2.06 %	Exploits	0.00 %	11.25 %
7	DrDoS_NetBIOS	0.00 %	4.14 %	Analysis	0.00 %	7.24 %
8	UDP-lag	0.00 %	0.36 %	Backdoor	0.00 %	6.59 %
9	Syn	0.00 %	7.66 %	Shellcode	0.00 %	4.29 %
10	DrDoS_DNS	0.00 %	3.75 %	Worms	0.00 %	0.48 %
11	DrDoS_SNMP	0.00 %	4.14 %	Total Count	12,970	35,168
12	DrDoS_SSDP	0.00 %	4.12 %			
13	DrDoS_NTP	0.00 %	3.59 %			
14	WebDDoS	0.00 %	0.04 %			
	Total Count	12,004	36,232			

Table 4
Sampling information for the IoT-IDAD-2024 dataset.

Index	IoT-IDAD-2024						
	Attack Category	TRAIN	TEST		Attack Category	TRAIN	TEST
1	Benign	53.50 %	54.90 %	9	DDoS-ACK_Fragmentation	0.00 %	4.80 %
2	DNS_Spoofing	7.23 %	2.22 %	10	DDoS-ICMP_Fragmentation	0.00 %	4.49 %
3	DictionaryBruteForce	7.19 %	3.37 %	11	XSS	0.00 %	3.83 %
4	DoS-SYN_Flood	7.18 %	4.73 %	12	DoS-UDP_Flood	0.00 %	3.37 %
5	Mirai-greeth_flood	7.18 %	3.26 %	13	DoS-HTTP_Flood	0.00 %	2.68 %
6	ArpSpoofing	7.18 %	1.87 %	14	SqlInjection	0.00 %	2.52 %
7	DDoS-HTTP_Flood	7.17 %	2.80 %	15	VulnerabilityScan	0.00 %	2.35 %
8	Uploading_Attack	3.38 %	0.63 %	16	DDoS-ICMP_Flood	0.00 %	2.17 %
Total Count						21,588	32,005

Table 5
IDS2017 comparison experiment.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
IR-IDS	**0.99**	**0.99**	**0.99**	**98.83 %**	**0.99**	**0.98**	**0.98**	**97.68 %**
KNN	0.77	0.73	0.75	75.40 %	0.86	0.77	0.71	80.18 %
SVM	0.83	0.77	0.79	79.21 %	0.79	0.78	0.78	78.02 %
CSA-RF	0.65	0.64	0.66	66.29 %	0.80	0.73	0.68	76.32 %
CGAN	0.87	0.86	0.86	86.16 %	0.94	0.95	0.97	95.56 %
CNN3D	0.89	0.89	0.89	88.68 %	0.94	0.94	0.94	94.21 %
DAE	0.65	0.63	0.66	66.14 %	0.73	0.74	0.74	74.53 %
AdaptIDS	0.83	0.81	0.84	83.58 %	0.94	0.87	0.80	88.21 %
SCAESVM	0.71	0.65	0.68	68.34 %	0.80	0.64	0.54	71.48 %
SNDAE	0.62	0.63	0.68	67.76 %	0.79	0.48	0.35	64.46 %
KAN	0.81	0.78	0.78	77.95 %	0.83	0.83	0.83	82.87 %
DT	0.56	0.49	0.60	59.84 %	0.65	0.58	0.61	61.26 %

Table 6
IDS2018 comparison experiment.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
IR-IDS	**0.99**	**0.99**	**0.99**	**99.37 %**	**0.99**	**0.99**	**0.99**	**99.48 %**
KNN	0.92	0.90	0.92	91.99 %	0.93	0.92	0.92	91.98 %
SVM	0.93	0.91	0.91	90.84 %	0.93	0.92	0.92	91.88 %
CSA-RF	0.86	0.8	0.83	82.83 %	0.80	0.75	0.76	75.86 %
CGAN	0.68	0.66	0.67	66.98 %	0.92	0.92	0.91	91.97 %
CNN3D	0.68	0.68	0.68	68.02 %	0.92	0.91	0.91	91.71 %
DAE	0.69	0.62	0.59	59.25 %	0.65	0.75	0.88	71.51 %
AdaptIDS	0.65	0.63	0.65	64.77 %	0.95	0.92	0.89	92.44 %
SCAESVM	0.63	0.61	0.60	60.28 %	0.78	0.84	0.92	83.67 %
SNDAE	0.65	0.63	0.64	64.37 %	0.65	0.40	0.28	58.20 %
KAN	0.93	0.91	0.90	90.45 %	0.84	0.81	0.81	80.99 %
DT	0.56	0.52	0.53	53.41 %	0.54	0.54	0.54	53.91 %

Table 7
DDOS2019 comparison experiment.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
IR-IDS	**1.00**	**1.00**	**1.00**	**99.57 %**	**1.00**	**1.00**	**1.00**	**99.79 %**
KNN	0.97	0.96	0.97	96.55 %	1.00	1.00	1.00	99.67 %
SVM	0.97	0.96	0.96	96.11 %	0.97	0.97	0.97	96.77 %
CSA-RF	0.75	0.79	0.85	84.82 %	0.90	0.88	0.88	87.91 %
CGAN	0.81	0.81	0.84	84.08 %	1.00	0.97	0.94	97.27 %
CNN3D	0.79	0.77	0.81	80.92 %	0.99	0.90	0.83	91.90 %
DAE	0.79	0.77	0.78	78.02 %	1.00	0.97	0.94	97.27 %
AdaptIDS	0.82	0.83	0.85	85.17 %	0.99	0.96	0.93	96.45 %
SCAESVM	0.84	0.81	0.83	82.60 %	1.00	0.97	0.95	97.35 %
SNDAE	0.72	0.64	0.67	67.02 %	0.85	0.71	0.60	77.43 %
KAN	0.96	0.94	0.95	94.61 %	0.98	0.98	0.98	97.82 %
DT	0.66	0.63	0.61	60.70 %	0.67	0.66	0.66	65.74 %

Table 8
UNSW-NB15 comparison experiment.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
IR-IDS	**1.00**	**1.00**	**1.00**	**99.52 %**	**0.99**	**0.99**	**0.99**	98.77 %
KNN	0.94	0.93	0.93	92.51 %	0.98	0.98	0.98	98.38 %
SVM	0.96	0.95	0.95	95.08 %	0.99	0.99	0.99	98.62 %
CSA-RF	0.93	0.88	0.88	87.87 %	0.99	0.99	0.99	**99.44 %**
CGAN	0.89	0.87	0.88	87.75 %	0.98	0.96	0.94	95.69 %
CNN3D	0.87	0.85	0.86	85.57 %	0.98	0.96	0.94	96.14 %
DAE	0.88	0.87	0.87	86.95 %	0.99	0.96	0.94	96.53 %
AdaptIDS	0.91	0.86	0.86	85.58 %	0.99	0.88	0.80	89.46 %
SCAESVM	0.88	0.86	0.86	86.16 %	0.99	0.94	0.90	94.56 %
SNDAE	0.71	0.66	0.65	65.30 %	0.70	0.57	0.49	63.82 %
KAN	0.94	0.93	0.92	92.41 %	0.99	0.99	0.99	98.58 %
DT	0.90	0.87	0.88	87.76 %	0.82	0.69	0.72	71.75 %

Table 9
IoT-IDAD-2024 comparison experiment.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
IR-IDS	**0.97**	**0.97**	**0.97**	**97.11 %**	**1.00**	**1.00**	**1.00**	**99.58 %**
KNN	0.82	0.76	0.74	73.97 %	0.83	0.81	0.80	80.46 %
SVM	0.83	0.73	0.71	70.89 %	0.78	0.78	0.78	77.69 %
CSA-RF	0.72	0.72	0.72	72.32 %	0.83	0.83	0.83	82.60 %
CGAN	0.83	0.82	0.83	82.82 %	0.84	0.83	0.83	83.15 %
CNN3D	0.81	0.78	0.80	79.91 %	0.83	0.83	0.83	83.06 %
DAE	0.77	0.74	0.74	73.79 %	0.63	0.62	0.62	61.54 %
AdaptIDS	0.77	0.57	0.51	50.84 %	0.79	0.76	0.76	76.48 %
SCAESVM	0.70	0.60	0.56	56.28 %	0.65	0.63	0.63	63.44 %
SNDAE	0.62	0.51	0.48	48.46 %	0.54	0.52	0.52	52.37 %
KAN	0.87	0.80	0.76	76.39 %	0.81	0.81	0.80	80.49 %
DT	0.49	0.33	0.28	27.70 %	0.48	0.47	0.47	47.08 %

leads to varying degrees of performance degradation, but the overall accuracy remains high. This demonstrates that the proposed method is highly modular, allowing it to integrate with other classification algorithms or feature selection methods to improve the performance of existing models.

From the overall accuracy perspective, after ablating the DMB-PS module, the algorithm's accuracy decreased by a maximum of 4.32 %, which validates the effectiveness of the DMB-PS algorithm. Further analysis of the detection accuracy changes for each type of attack reveals that DMB-PS shows significant performance improvements in identifying specific unknown attacks. For example, the "Bot" attack in the IDS2017 dataset (accuracy dropped by 98.03 %) and the "VulnerabilityScan" attack in the IoT-IDAD-2024 dataset (accuracy dropped by 30.59 %) demonstrate that the feature selection mechanism significantly enhances the algorithm's predictive capability. This improvement mainly results from the feature selection process, which reduces the number of nodes that the Shapley Tree needs to evaluate, thereby minimizing the interference of redundant noise and optimizing the model's prediction performance. Although, in certain specific cases, feature selection may remove features that are highly discriminative for certain attacks, leading to unchanged or slightly improved detection accuracy for those attacks when using the full feature set, the overall accuracy still shows a stable upward trend, fully demonstrating the substantial contribution of DMB-PS to the IR-IDS system.

In the comparison experiment where Shapley Tree was replaced with the traditional Decision Tree (DT), the overall accuracy showed significant improvement over using only DT, indicating that DMB-PS and KAN—CVAE have good algorithm compatibility and can effectively integrate with various classification algorithms. This integration improves the algorithm's accuracy without introducing black-box models. When using Shapley Tree alone for classification, the classification accuracy remained stable between 92.32 % and 98.76 %. This

Table 10
IDS2017 ablation experiments.

DMB-PS	√	×	√	√	×
Shapley-Tree	√	√	DT	√	√
KAN—CVAE	√	√	√	×	×
Overall Accuracy	98.83 %	97.56 %	95.91 %	96.37 %	95.06 %
BENIGN	98.77 %	99.14 %	98.58 %	98.77 %	99.14 %
DoS Slowhttptest	99.83 %	98.26 %	98.84 %	99.83 %	98.26 %
DDoS	98.79 %	99.13 %	98.86 %	98.79 %	99.13 %
FTP-Patator	100 %	99.93 %	100 %	100 %	99.93 %
SSH-Patator	100 %	99.93 %	100 %	100 %	99.93 %
DoS slowloris	99.80 %	99.80 %	99.80 %	99.80 %	99.80 %
PortScan	99.87 %	99.87 %	99.87 %	99.87 %	99.87 %
DoS Hulk	98.26 %	97.59 %	98.20 %	98.26 %	97.59 %
Web Attack Brute Force	96.88 %	98.44 %	96.88 %	96.88 %	98.44 %
Bot	98.82 %	0.79 %	96.84 %	96.84 %	0.39 %
DoS GoldenEye	99.81 %	99.88 %	63.83 %	68.14 %	67.83 %
Infiltration	2.78 %	30.56 %	2.78 %	2.78 %	2.78 %
Web Attack XSS	93.56 %	93.25 %	93.56 %	93.56 %	93.25 %
Web Attack Sql Injection	23.81 %	42.86 %	19.05 %	19.05 %	38.10 %
Heartbleed	90.91 %	100 %	90.91 %	90.91 %	100 %

Table 11
IDS2018 ablation experiments.

DMB-PS	√	×	√	√	×
Shapley-Tree	√	√	DT	√	√
KAN—CVAE	√	√	√	×	×
Overall Accuracy	99.37 %	96.66 %	97.83 %	97.91 %	95.26 %
Benign	99.33 %	96.93 %	99.22 %	99.33 %	96.93 %
SSH-Bruteforce	100 %	100 %	100 %	100 %	100 %
DoS attacks-GoldenEye	99.53 %	99.42 %	99.53 %	99.53 %	99.42 %
DoS attacks-SlowHTTPTest	100 %	100 %	100 %	100 %	100 %
DDOS attack-HOIC	99.65 %	99.86 %	99.79 %	99.65 %	99.86 %
Infilteration	98.51 %	98.62 %	98.51 %	98.51 %	98.62 %
Bot	97.27 %	96.87 %	97.33 %	97.27 %	96.87 %
DDoS attacks-LOIC—HTTP	100 %	100 %	100 %	100 %	100 %
DoS attacks-Hulk	100 %	99.26 %	99.26 %	100 %	99.26 %
DoS attacks-Slowloris	99.53 %	65.68 %	37.60 %	37.60 %	5.93 %
FTP-BruteForce	97.64 %	49.61 %	97.64 %	96.85 %	48.03 %
Brute Force -Web	100 %	100 %	100 %	100 %	100 %
Brute Force -XSS	97.39 %	18.26 %	97.39 %	96.52 %	17.39 %
SQL Injection	83.33 %	16.67 %	69.44 %	69.44 %	16.67 %
DDOS attack-LOIC-UDP	100 %	100 %	100 %	100 %	100 %

Table 12
DDOS2019 ablation experiments.

DMB-PS	√	×	√	√	×
Shapley-Tree	√	√	DT	√	√
KAN—CVAE	√	√	√	×	×
Overall Accuracy	99.57 %	99.43 %	99.49 %	99.48 %	97.59 %
BENIGN	99.93 %	99.88 %	99.94 %	99.93 %	99.88 %
DrDoS_UDP	99.16 %	99.16 %	99.16 %	99.16 %	99.16 %
DrDoS_LDAP	94.60 %	94.13 %	94.60 %	94.60 %	94.13 %
Portmap	100 %	100 %	100 %	100 %	100 %
TFTP	99.80 %	99.53 %	99.80 %	99.80 %	99.53 %
DrDoS_MSSQL	98.80 %	98.53 %	98.93 %	98.80 %	98.53 %
DrDoS_NetBIOS	100 %	100 %	98.60 %	98.60 %	79.40 %
UDP-lag	100 %	97.71 %	99.24 %	99.24 %	97.71 %
Syn	99.89 %	99.50 %	99.53 %	99.50 %	99.50 %
DrDoS_DNS	99.71 %	99.48 %	99.71 %	99.71 %	99.48 %
DrDoS_SNMP	100 %	100 %	100 %	100 %	76.38 %
DrDoS_SSDP	98.73 %	98.19 %	98.73 %	98.73 %	98.19 %
DrDoS_NTP	99.85 %	99.77 %	99.85 %	99.85 %	99.62 %
WebDDoS	7.69 %	7.69 %	7.69 %	7.69 %	7.69 %

Table 13
UNSW-NB15 ablation experiments.

DMB-PS	√	×	√	√	×
Shapley-Tree	√	√	DT	√	√
KAN—CVAE	√	√	√	×	×
Overall Accuracy	99.52 %	98.79 %	99.16 %	99.37 %	98.76 %
Normal	99.57 %	99.45 %	99.20 %	99.57 %	99.45 %
Fuzzers	97.88 %	91.71 %	97.33 %	97.88 %	91.71 %
Generic	99.69 %	98.65 %	96.48 %	99.69 %	98.65 %
DoS	97.31 %	97.31 %	98.45 %	97.31 %	97.31 %
Reconnaissance	99.27 %	94.63 %	99.08 %	99.27 %	94.63 %
Exploits	99.87 %	99.85 %	99.90 %	99.85 %	99.77 %
Analysis	99.84 %	99.92 %	99.84 %	98.23 %	99.92 %
Backdoor	99.83 %	99.74 %	99.78 %	99.57 %	99.74 %
Shellcode	99.87 %	99.80 %	100 %	99.87 %	99.80 %
Worms	100 %	100 %	100 %	100 %	100 %

performance is still superior to the DT-only method in Section 4.2 and further confirms the significant advantages of Shapley Tree in improving model accuracy and robustness. It is worth noting that for certain specific attacks, the classification paths of Decision Tree (DT) and Shapley Tree (ST) are similar, resulting in unchanged detection accuracy before and after replacing the algorithm. Even if DT shows better classification performance in some cases, the accuracy for these attacks may improve. However, overall, the cases with accuracy degradation are more common, such as in the "DoS-UDP_Flood" attack in the IoT-IDAD-2024 dataset (decreased by 16.22 %) and the "SQL Injection" attack in the IDS2018 dataset (decreased by 13.89 %). In these cases, the tree model optimized with Shapley values demonstrated stronger detection capability for unknown attacks.

In the ablation experiment of the KAN—CVAE module, the detection accuracy for known attacks remained stable, but the identification accuracy for some unknown attacks significantly decreased. For example, the "DoS GoldenEye" attack in the IDS2017 dataset dropped by 31.67 %, and the "DDoS-ACK_Fragmentation" attack in the IoT-IDAD-2024 dataset dropped by 5.37 %. This phenomenon suggests that the reconstruction error features play a crucial role in improving the model's ability to discriminate unknown attacks. Although Shapley Tree itself has strong classification capabilities and can classify some unknown attack types as non-benign traffic during the pre-classification phase, these unknown attacks are still accurately identified as attack categories after further classification by KAN—CVAE, resulting in minimal fluctuation in their detection accuracy. However, after ablating the KAN—CVAE module, the overall accuracy of the model showed a significant decline, which further validates the effectiveness of the KAN—CVAE module in enhancing the model's performance.

Table 14
IoT-IDAD-2024 ablation experiments.

DMB-PS	√	×	√	√	×
Shapley-Tree	√	√	DT	√	√
KAN—CVAE	√	√	√	×	×
Overall Accuracy	97.11 %	92.79 %	96.14 %	95.87 %	92.32 %
Benign	99.29 %	96.06 %	99.25 %	99.29 %	96.06 %
DNS_Spoofing	97.61 %	92.55 %	97.89 %	97.61 %	92.55 %
DictionaryBruteForce	99.26 %	89.35 %	99.54 %	99.26 %	89.35 %
DoS-SYN_Flood	99.47 %	99.01 %	99.47 %	99.47 %	99.01 %
Mirai-greeth_flood	97.99 %	90.33 %	98.08 %	97.99 %	90.33 %
ArpSpoofing	96.65 %	85.09 %	96.65 %	96.65 %	85.09 %
DDoS-HTTP_Flood	98.88 %	97.10 %	97.21 %	98.88 %	97.10 %
Uploading_Attack	97.03 %	90.10 %	97.03 %	97.03 %	90.10 %
DDoS-ACK_Fragmentation	98.18 %	97.27 %	92.51 %	92.45 %	91.34 %
DDoS-ICMP_Fragmentation	93.74 %	92.28 %	93.39 %	93.53 %	92.28 %
XSS	86.88 %	83.94 %	86.47 %	84.19 %	83.70 %
DoS-UDP_Flood	95.18 %	87.67 %	78.96 %	78.68 %	87.30 %
DoS-HTTP_Flood	86.71 %	95.45 %	84.73 %	78.67 %	94.76 %
SqlInjection	96.15 %	81.39 %	95.78 %	96.15 %	79.90 %
VulnerabilityScan	81.65 %	51.06 %	81.25 %	79.52 %	48.54 %
DDoS-ICMP_Flood	86.17 %	77.67 %	86.17 %	85.01 %	75.36 %

4.4. Feature selection comparison

Next, we conduct comparative experiments on both the feature selection module and the classification module, with the comparison results focused on the multi-class prediction accuracy.

In this section, we compare the proposed causal feature selection method, DMB-PS, with existing methods. To assess the effectiveness of the propensity score-based conditional testing method (PS), we compare it with mutual information-based (MI) and G^2-test-based (G^2) methods. The threshold for the basic conditional test is set at 0.1. To ensure valid comparison results, we apply the K-means method for preprocessing before using the testing methods, ensuring consistency across all preprocessing steps.

To validate the effectiveness of the proposed DMB algorithm in searching for the Markov blanket, we compare it with the advanced Markov blanket search algorithm, BAMB (Ling et al., 2019). For evaluating the performance of our method in anomaly detection, we compare it with advanced feature selection methods like ELIDS (Fatima et al., 2024) and CSA (Mohiuddin et al., 2023). Since CSA-RF is also a filter-based method, we separate its feature selection and classification algorithms for comparison. The parameters used in the comparison include feature classification time, the number of selected features, and prediction accuracy. The classification algorithms compared include the baseline models, DT and KAN, as well as advanced models like RF (Mohiuddin et al., 2023), KNN (Pajouh et al., 2019) and SVM (Jeong et al., 2022). Given the inclusion of Shapley Tree and KAN—CVAE, we refer to the classification algorithm in this paper as SHAP-CVAE. The experimental results are presented in Tables 15–19. "Nothing" indicates no feature selection is performed, and the time is measured in seconds. The feature selection results for the four datasets are shown in Table 20.

From the comparison with BAMB, it is evident that the proposed method improves the search efficiency of the Markov blanket in all three conditional testing methods. It also identifies more accurate feature subsets, significantly improving the accuracy of various classifiers, demonstrating the effectiveness of the DMB approach.

When compared to advanced methods like ELIDS, which uses traditional correlation-based approaches, our method requires more time but selects fewer features, effectively enhancing the accuracy of various classification algorithms. In contrast, CSA, which involves repetitive adjustments through PSO during feature selection, requires less time but often leads to lower accuracy. This is because CSA only performs feature selection on the training set, and the selected features are often insufficient for discriminating unknown attacks, resulting in reduced accuracy.

In the comparison of conditional testing methods, our method proves more efficient than the mutual information approach across all five datasets. Furthermore, by incorporating causal relationships based on causal effects, our algorithm effectively mitigates the impact of high-dimensional features. In contrast, traditional G^2 and mutual information methods, which suffer from the curse of dimensionality, select fewer relevant features, significantly reducing accuracy.

4.5. Classification algorithm comparison

Finally, the effectiveness of the classification algorithms was compared. To ensure the validity of the results, the feature selection process for all classification algorithms was replaced with DMB-PS, and the classification algorithm proposed in this paper is named SHAP-CVAE. The comparison was conducted for both multi-class and binary classification tasks, with evaluation metrics including Precision, F1, Recall, and Accuracy. The results are presented in Tables 21–25, where the highest value in each column is highlighted in bold.

When comparing the results in Tables 21–25 with those in Tables 5–9, it is evident that after applying feature selection with the DMB-PS algorithm, the accuracy of all classification algorithms in the multi-class setting has generally improved. This demonstrates the effectiveness and versatility of our feature selection method. However, in the binary classification case, some algorithms showed a decrease in accuracy, likely due to the reduced number of features, which can lead to overfitting.

From the classification results, both in binary and multi-class tasks,

Table 15
Feature selection method comparison results for IDS2017.

Methods	Time(s)	Number	DT	KAN	RF	SVM	KNN	SHAP-CVAE
Nothing	0	77	59.84 %	77.95 %	66.29 %	79.21 %	75.40 %	97.56 %
ELIDS	9.95	65	65.04 %	79.45 %	59.8 %	88.18 %	87.18 %	97.64 %
CSA	388.64	23	54.36 %	84.15 %	66.29 %	86.9 %	84.54 %	95.58 %
BAMB-G2	38.83	3	68.12 %	73.58 %	66.35 %	76.94 %	75.01 %	92.22 %
BAMB-MI	361.04	5	63.53 %	76.61 %	68.27 %	74.91 %	77.66 %	88.97 %
BAMB-PS	21,029.00	11	60.91 %	66.63 %	64 %	67.64 %	68.7 %	90.01 %
DMB-G2	27.83	3	67.61 %	65.29 %	64.31 %	68.37 %	71.11 %	89.77 %
DMB-MI	73.95	5	63.53 %	78.21 %	68.26 %	74.91 %	77.66 %	89.65 %
DMB-PS	37.46	18	**78.12 %**	**95.13 %**	**93.08 %**	**96.23 %**	**92.96 %**	**98.83 %**

Table 16
Feature selection method comparison results for IDS2018.

Methods	Time(s)	Number	DT	KAN	RF	SVM	KNN	SHAP-CVAE
Nothing	0	76	53.41 %	90.45 %	72.00 %	90.84 %	91.99 %	96.66 %
ELIDS	9.79	68	**69.94 %**	89.99 %	84.48 %	91.64 %	91.99 %	96.9 %
CSA	610.16	50	68.41 %	90.8 %	82.83 %	91.55 %	**93.16 %**	96.21 %
BAMB-G2	52.32	3	48.18 %	84.95 %	51.73 %	85.24 %	48.48 %	94.93 %
BAMB-MI	319.08	5	53.75 %	82.98 %	58.76 %	90.54 %	76.09 %	97.08 %
BAMB-PS	538.83	9	63.02 %	93.8 %	67.95 %	91.23 %	91.95 %	96.69 %
DMB-G2	25.20	2	47.25 %	67.65 %	51.02 %	68.11 %	55.59 %	72.16 %
DMB-MI	91.99	4	52.29 %	76.67 %	50.45 %	87.8 %	64.53 %	97.13 %
DMB-PS	39.46	21	68.50 %	**95.35 %**	**73.63 %**	**93.08 %**	92.02 %	**99.37 %**

Table 17
Feature selection method comparison results for DDOS2019.

Methods	Time(s)	Number	DT	KAN	RF	SVM	KNN	SHAP-CVAE
Nothing	0	76	60.70 %	94.61 %	84.82 %	96.11 %	96.55 %	99.43 %
ELIDS	5.85	75	60.83 %	**94.99 %**	84.88 %	95.22 %	96.55 %	98.82 %
CSA	90.86	34	83.55 %	93.78 %	84.82 %	90.77 %	94.33 %	98.03 %
BAMB-G2	9.38	2	**90.23 %**	85.89 %	90.66 %	89.8 %	85.58 %	94.57 %
BAMB-MI	97.36	4	87.62 %	92.37 %	88.74 %	93.55 %	95.32 %	98.34 %
BAMB-PS	3339.94	11	72.32 %	89.81 %	82.82 %	92.89 %	93.61 %	98.36 %
DMB-G2	16.62	2	69.61 %	78.14 %	74.24 %	85.12 %	80.64 %	94.23 %
DMB-MI	23.35	4	79.69 %	94.1 %	88.37 %	93.55 %	95.32 %	97.96 %
DMB-PS	22.00	19	62.19 %	94.90 %	**91.94 %**	**98.00 %**	**96.67 %**	**99.57 %**

SHAP-CVAE consistently performs at a higher level than other classification algorithms. In the multi-class classification, it improves accuracy by at least 3.7 % compared to other methods, proving the effectiveness of the proposed classification algorithm.

5. Discussion

5.1. Interpretability of IR-IDS

The interpretability discussed in this paper primarily refers to model interpretability, specifically the ability to explain model decisions and predictions in a way that is easily understood by humans (Lipton, 2018). This understandable decision-making process provides a trustworthy inference mechanism for IDS (Pai et al., 2024), making it crucial for IDS. Model interpretability is mainly demonstrated through the transparency of the model and the mathematical provability of its decision-making process. The IR-IDS method proposed in this paper achieves model interpretability in two key stages.

In the causal feature selection phase, we employ the propensity score-based DMB-PS algorithm, which is grounded in causal inference theory. Unlike black-box feature selection methods, our approach identifies causal relationships between features and the target variable through propensity score-based conditional testing. This probabilistic inference method ensures all calculation processes are traceable and statistically verifiable, providing clear justification for feature selection decisions. The Double Markov Blanket (DMB) algorithm further en-

Table 18
Feature selection method comparison results for UNSW-NB15.

Methods	Time(s)	Number	DT	KAN	RF	SVM	KNN	SHAP-CVAE
Nothing	0	39	87.76 %	92.41 %	87.87 %	95.08 %	92.51 %	98.79 %
ELIDS	2.80	37	87.65 %	92.41 %	88.95 %	94.6 %	92.5 %	98.22 %
CSA	164.71	25	76.86 %	88.24 %	87.87 %	93.06 %	90.42 %	97.6 %
BAMB-G2	20.02	3	84.88 %	91.19 %	85.09 %	95.66 %	90.9 %	97.81 %
BAMB-MI	36.65	2	90.15 %	91.17 %	89.92 %	92.79 %	89.29 %	89.21 %
BAMB-PS	3396.80	11	75.18 %	89.86 %	**94.75 %**	91.75 %	88.85 %	94.96 %
DMB-G2	12.13	1	74.91 %	82.31 %	74.91 %	82.31 %	74.33 %	82.69 %
DMB-MI	6.88	3	**90.2 %**	89.68 %	90.12 %	91.57 %	84.75 %	89.61 %
DMB-PS	13.33	18	78.12 %	**95.13 %**	93.08 %	**96.23 %**	**92.96 %**	**99.52 %**

Table 19
Feature selection method comparison results for IoT-IDAD-2024.

Methods	Time(s)	Number	DT	KAN	RF	SVM	KNN	SHAP-CVAE
Nothing	0.00	79	27.70 %	76.39 %	56.46 %	70.89 %	73.97 %	92.79 %
ELIDS	12.74	70	44.16 %	**82.91 %**	52.20 %	77.48 %	81.72 %	93.96 %
CSA	599.37	35	57.98 %	75.63 %	72.32 %	75.53 %	70.31 %	92.57 %
BAMB-G2	44.39	2	45.02 %	39.15 %	45.21 %	62.03 %	39.59 %	77.27 %
BAMB-MI	228.19	2	70.40 %	70.17 %	70.40 %	70.26 %	55.71 %	70.40 %
BAMB-PS	1464.76	9	31.20 %	63.69 %	69.68 %	74.71 %	64.14 %	90.85 %
DMB-G2	21.87	2	45.02 %	49.84 %	45.29 %	62.03 %	39.59 %	77.27 %
DMB-MI	43.90	2	60.68 %	70.17 %	70.40 %	70.26 %	55.71 %	70.40 %
DMB-PS	45.41	22	**63.32 %**	76.89 %	**76.30 %**	**78.62 %**	**83.02 %**	**97.11 %**

Table 20
DMB-PS feature selection results.

Dataset	Results of Feature Selection
IDS2017	Fwd Packet Length Min, Fwd Packet Length Mean, Bwd Packet Length Max, Bwd Packet Length Mean, Bwd Packet Length Std, Flow IAT Mean, Flow IAT Std, Flow IAT Max, Fwd IAT Total, Fwd IAT Std, Bwd IAT Total, Bwd IAT Mean, Bwd IAT Std, Fwd PSH Flags, PSH Flag Count, Down/Up Ratio, Average Packet Size, Init_Win_bytes_forward
IDS2018	Fwd Pkt Len Min, Bwd Pkt Len Mean, Bwd Pkt Len Std, Flow Pkts/s, Flow IAT Std, Fwd IAT Tot, Fwd IAT Max, Bwd IAT Tot, Bwd IAT Std, Pkt Len Max, Pkt Len Std, Pkt Len Var, RST Flag Cnt, PSH Flag Cnt, Pkt Size Avg, Fwd Seg Size Avg, Init Fwd Win Byts, Init Bwd Win Byts, Fwd Seg Size Min, Active Mean, Idle Mean
DDOS2019	Flow Duration, Total Fwd Packets, Fwd Packet Length Max, Fwd Packet Length Min, Bwd Packet Length Min, Bwd Packet Length Std, Flow Bytes/s, Fwd IAT Mean, Fwd Header Length, Fwd Packets/s, Min Packet Length, Packet Length Mean, SYN Flag Count, ACK Flag Count, URG Flag Count, Init_Win_bytes_forward, Init_Win_bytes_backward, act_data_pkt_fwd, Active Mean
UNSW-NB15	sload, sinpkt, sjit, dwin, tcprtt, synack, ackdat, smean, dmean, ct_srv_src, ct_state_ttl, ct_dst_ltm, ct_src_dport_ltm, ct_dst_sport_ltm, ct_dst_src_ltm, ct_flw_http_mthd, ct_src_ltm, ct_srv_dst
IoT-IDAD-2024	Bwd Packet Length Max, Bwd Packet Length Min, Bwd Packet Length Mean, Flow IAT Std, Fwd IAT Total, Fwd IAT Mean, Fwd IAT Std, Fwd IAT Max, Fwd IAT Min, Packet Length Min, Packet Length Max, Packet Length Mean, FIN Flag Count, SYN Flag Count, RST Flag Count, Down/Up Ratio, Fwd Segment Size Avg, FWD Init Win Bytes, Bwd Init Win Bytes, Fwd Seg Size Min, Active Max, Active Min

hances interpretability by identifying the minimal feature subset containing all necessary information to predict the target variable. This transparent process follows well-established causal theory principles. When a feature is included in the Markov Blanket, users can understand that this feature either directly causes the target variable, is directly caused by it, or shares a common cause with it. This causal explanation provides a level of interpretability that correlation-based methods cannot achieve. The mathematical theory of this phase has been effectively proven in Section 3.2.

In the robust anomaly classification phase, the Shapley Tree model offers excellent interpretability by quantifying each feature's contribution to classification decisions using Shapley values. Unlike traditional decision trees that rely solely on information gain, our approach provides fair attribution of feature importance based on game theory principles. These clear attributions help us not only understand what predictions the model makes, but why it makes them. Additionally, the decision tree model itself is a transparent and interpretable model structure, and the Shapley Tree model is interpretable as well. The KAN—CVAE component, while leveraging deep learning techniques, maintains interpretability through the integration of Kolmogorov-Arnold Networks. Traditional neural networks are often criticized as "black boxes," but KAN, based on Kolmogorov's superposition theorem, decomposes complex multi-dimensional functions into simpler one-dimensional components. This decomposition makes internal representations more transparent and allows better understanding of latent space transformations, providing insight into how unknown attacks are detected based on reconstruction errors. The mathematical proof of KAN's interpretability has been thoroughly derived (Liu et al., 2025), and to ensure its interpretable characteristics, we configured only one hidden layer with 12 nodes, further guaranteeing the interpretability of its structure.

Overall, IR-IDS achieves model interpretability, enabling security professionals to understand why certain traffic patterns are flagged as anomalous and facilitating more informed responses to potential threats.

Table 21
Comparison of classification algorithms on IDS2017.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
SHAP-CVAE	**0.99**	**0.99**	**0.99**	**98.83 %**	**0.99**	**0.98**	**0.98**	**97.68 %**
KNN	0.94	0.93	0.93	92.96 %	0.88	0.88	0.88	88.05 %
SVM	0.97	0.96	0.96	96.23 %	0.79	0.81	0.82	81.18 %
RF	0.96	0.93	0.93	93.08 %	0.81	0.84	0.87	84.17 %
CGAN	0.91	0.91	0.91	90.79 %	0.96	0.96	0.96	96.44 %
CNN3D	0.91	0.91	0.91	90.78 %	0.96	0.96	0.96	96.22 %
DAE	0.75	0.70	0.72	72.21 %	0.89	0.80	0.72	82.37 %
AdaptIDS	0.90	0.90	0.90	89.85 %	0.97	0.95	0.94	95.38 %
SCAESVM	0.76	0.69	0.73	72.54 %	0.96	0.86	0.78	87.82 %
SNDAE	0.72	0.7	0.73	72.84 %	0.80	0.82	0.85	82.49 %
KAN	0.96	0.95	0.95	95.13 %	0.90	0.89	0.89	89.17 %
DT	0.89	0.82	0.78	78.12 %	0.81	0.73	0.74	74.40 %

Table 22
Comparison of classification algorithms on IDS2018.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
SHAP-CVAE	**0.99**	**0.99**	**0.99**	**99.37 %**	**0.99**	**0.99**	**0.99**	**99.48 %**
KNN	0.92	0.90	0.92	92.02 %	0.93	0.92	0.92	92.03 %
SVM	0.94	0.92	0.93	93.08 %	0.93	0.92	0.92	92.05 %
RF	0.64	0.65	0.74	73.63 %	0.84	0.77	0.78	78.18 %
CGAN	0.72	0.71	0.71	71.49 %	0.94	0.92	0.90	92.55 %
CNN3D	0.72	0.72	0.72	71.81 %	0.95	0.92	0.89	92.51 %
DAE	0.72	0.71	0.72	71.80 %	0.96	0.91	0.87	92.08 %
AdaptIDS	0.70	0.69	0.71	70.83 %	0.95	0.94	0.93	94.09 %
SCAESVM	0.68	0.66	0.67	66.95 %	0.96	0.91	0.87	91.77 %
SNDAE	0.71	0.67	0.7	69.74 %	0.78	0.70	0.63	73.96 %
KAN	0.95	0.95	0.95	95.35 %	0.96	0.95	0.95	95.46 %
DT	0.64	0.64	0.68	68.50 %	0.70	0.70	0.70	70.32 %

Table 23
Comparison of classification algorithms on DDOS2019.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
SHAP-CVAE	**1.00**	**1.00**	**1.00**	**99.57 %**	**1.00**	**1.00**	**1.00**	**99.79 %**
KNN	0.97	0.96	0.97	96.67 %	1.00	1.00	1.00	99.69 %
SVM	0.98	0.98	0.98	98.00 %	0.98	0.98	0.98	97.99 %
RF	0.91	0.91	0.92	91.94 %	0.92	0.91	0.91	90.86 %
CGAN	0.84	0.86	0.88	88.17 %	0.99	0.97	0.95	97.23 %
CNN3D	0.85	0.86	0.88	88.18 %	0.99	0.99	0.98	98.85 %
DAE	0.85	0.86	0.88	88.32 %	0.99	0.98	0.96	97.91 %
AdaptIDS	0.85	0.86	0.88	88.13 %	1.00	0.99	0.99	98.81 %
SCAESVM	0.96	0.94	0.95	94.63 %	0.98	0.99	0.98	99.13 %
SNDAE	0.83	0.80	0.81	80.67 %	1.00	0.82	0.80	84.43 %
KAN	0.95	0.94	0.95	94.90 %	0.98	0.98	0.98	97.89 %
DT	0.75	0.62	0.62	62.19 %	0.85	0.77	0.77	77.50 %

Table 24
Comparison of classification algorithms on UNSW-NB15.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
SHAP-CVAE	**1.00**	**1.00**	**1.00**	**99.52 %**	**0.99**	**0.99**	**0.99**	98.77 %
KNN	0.94	0.93	0.93	92.96 %	0.98	0.98	0.98	98.48 %
SVM	0.97	0.96	0.96	96.23 %	0.99	0.99	0.99	99.42 %
RF	0.96	0.93	0.93	93.08 %	0.99	0.99	0.99	**99.49 %**
CGAN	0.94	0.93	0.93	93.15 %	0.98	0.98	0.98	97.77 %
CNN3D	0.94	0.93	0.93	92.54 %	0.98	0.98	0.98	97.58 %
DAE	0.93	0.91	0.91	90.96 %	0.98	0.98	0.98	97.93 %
AdaptIDS	0.90	0.90	0.90	90.12 %	1.00	0.96	0.93	96.41 %
SCAESVM	0.91	0.90	0.90	90.46 %	0.98	0.98	0.97	97.74 %
SNDAE	0.84	0.82	0.80	80.39 %	0.81	0.85	0.89	84.08 %
KAN	0.96	0.95	0.95	95.13 %	0.99	0.99	0.99	98.90 %
DT	0.89	0.82	0.78	78.12 %	0.83	0.79	0.80	79.69 %

5.2. Robustness of IR-IDS

Robustness in this paper refers to the ability to maintain reliable performance under various conditions, particularly when confronted with previously unseen or unknown attack patterns (Nguyen and Le, 2023). A robust IDS should be resilient against adversarial techniques and maintain high detection accuracy across diverse network environments and evolving threat landscapes.

The robustness of IR-IDS is validated by the experimental results in Section 4. In our experiments, we ensured that each test set contained 11.26 % to 29.85 % of attacks not present in the training set. The results demonstrate that our method can effectively identify unknown attacks as anomalies, with an accuracy reaching as high as 97.11 %. Compared to advanced methods, IR-IDS maintains performance advantages in both multi-class and binary classification tasks. This consistent superiority across different network contexts and attack scenarios provides strong evidence of the method's robustness and its ability to generalize to real-world situations.

IR-IDS ensures robustness primarily through two mechanisms: causal feature selection and KAN—CVAE integration.

The causal feature selection process filters out irrelevant features, reducing noise in the data and helping the model focus on features with genuine predictive power, thereby enhancing the model's generalization capability. By ensuring the feature space is both efficient and relevant, IR-IDS can better handle the complexity of network traffic data and improve its ability to detect unknown attack types. Consequently, this feature selection method provides IR-IDS with strong robustness, particularly in effectively detecting new types of attacks. As seen in the results from Section 4.3, when the DMB-PS method was ablated, the unknown attack "Bot" in the IDS2017 dataset caused the accuracy to drop by 98.03 %, and the unknown attack "VulnerabilityScan" in the IoT-IDAD-2024 dataset caused a drop of 30.59 %. These results clearly

Table 25
Comparison of classification algorithms on IoT-IDAD-2024.

Method	Multiclass Classification				Binary Classification			
	Precision	F1	Recall	Accuracy	Precision	F1	Recall	Accuracy
SHAP-CVAE	**0.97**	**0.97**	**0.97**	**97.11 %**	**1.00**	**1.00**	**1.00**	**99.58 %**
KNN	0.85	0.82	0.83	83.02 %	0.87	0.86	0.86	86.26 %
SVM	0.81	0.75	0.79	78.62 %	0.83	0.80	0.80	80.36 %
RF	0.79	0.76	0.76	76.30 %	0.87	0.87	0.87	86.55 %
CGAN	0.80	0.80	0.81	81.18 %	0.83	0.83	0.83	82.79 %
CNN3D	0.84	0.83	0.83	83.39 %	0.83	0.83	0.83	83.27 %
DAE	0.69	0.56	0.53	52.69 %	0.74	0.74	0.74	73.97 %
AdaptIDS	0.86	0.85	0.85	85.37 %	0.87	0.87	0.87	86.55 %
SCAESVM	0.75	0.68	0.66	65.52 %	0.70	0.69	0.69	68.88 %
SNDAE	0.72	0.64	0.62	62.22 %	0.72	0.66	0.67	66.81 %
KAN	0.84	0.79	0.77	76.89 %	0.84	0.81	0.81	80.92 %
DT	0.80	0.69	0.63	63.32 %	0.72	0.60	0.62	62.45 %

demonstrate that the feature selection mechanism significantly enhances IR-IDS's ability to predict unknown attacks, thus improving the system's robustness.

Furthermore, the combination of KAN—CVAE also effectively enhances IR-IDS's robustness. KAN contributes by efficiently representing non-linear relationships, helping the model accurately identify unknown attack types. CVAE, through its reconstruction error mechanism, aids the model in discovering traffic patterns that differ from the training data, further improving its ability to detect unknown attacks. The integration of KAN and CVAE enables IR-IDS to maintain high detection capabilities even when facing new, previously unseen attacks. As shown in the results from Section 4.3, when KAN—CVAE was ablated, the accuracy for the unknown attack "DoS GoldenEye" in the IDS2017 dataset dropped by 31.67 %, and the unknown attack "DDoS-ACK_Fragmentation" in the IoT-IDAD-2024 dataset dropped by 5.37 %. These results effectively demonstrate KAN—CVAE's ability to detect unknown attacks and highlight its contribution to IR-IDS's robustness.

Overall, from the four sets of experimental results in Section 4, we can observe the excellent performance of IR-IDS in detecting previously unseen or unknown attack patterns, which effectively demonstrates the strong detection robustness of IR-IDS in handling diverse and evolving attack environments.

5.3. Future work

Although our method performs excellently in a laboratory environment and has been extensively evaluated on five different datasets, to make IR-IDS truly effective, we must consider how to apply it to real-world Intrusion Detection Systems (IDS). During actual deployment, the performance and effectiveness of an IDS are often influenced by multiple factors. Therefore, the next step is to explore how to implement and deploy IR-IDS in the real world.

Firstly, the architecture of IR-IDS is modular and highly flexible, which makes it easy to integrate with existing network defense systems. IR-IDS can be deployed as a bypass monitoring solution or directly integrated into the network infrastructure. For bypass deployment, IR-IDS can analyze mirrored network traffic without interfering with the main data flow, minimizing the potential impact on network performance. For direct integration, the system can be placed at key network nodes to provide real-time traffic analysis and threat detection.

Regarding hardware requirements, our experiments show that the IR-IDS method can operate effectively on standard server hardware. The minimum recommended specifications include an Intel(R) Xeon(R) W-2225 processor with a clock speed of 4.1 GHz and 32 GB of memory. Due to the relatively lightweight neural network structure in our method, with a limited number of layers and nodes, it does not require specialized GPU hardware for operation and deployment.

Additionally, IR-IDS must have high real-time capabilities and responsiveness in practical deployment. To ensure this, we plan to first collect a certain amount of data for model pre-training, then deploy the trained model in real-world settings. To enhance the model's capability, we will summarize data traffic weekly and adjust and optimize the IR-IDS model accordingly, ensuring it effectively adapts to new attack patterns and changes in network traffic.

In conclusion, in the future, we will further consider the practical deployment capabilities of IR-IDS, address the technical and implementation challenges in the real world, and optimize its performance in real-world applications to ensure its stable operation and ability to address various complex network security threats.

6. Conclusion

This paper presents IR-IDS, a novel and robust intrusion detection method that comprises two main stages: causal feature selection and robust anomaly classification. In the causal feature selection phase, we propose a causal-effect-based conditional testing method, which efficiently searches the Markov blankets of the target variable and selects the optimal feature subset. Building upon this, we introduce Shapley Tree, a transparent classification model that combines the TreeExplainer model with decision trees, enabling fine-grained classification of attacks. Finally, we propose KAN—CVAE, an interpretable model capable of detecting anomalies from unknown attacks, based on Extreme Value Theory (EVT) and classification results. Experimental results on five datasets show that the proposed method not only enhances the multi-classification performance of intrusion detection and ensures model interpretability but also effectively adapts to scenarios involving unknown attacks.

In the future, we will further consider the challenges related to real-world deployment.

CRediT authorship contribution statement

Yazhuo Gao: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Lin Yang:** Writing – review & editing, Software, Project administration, Investigation, Funding acquisition, Conceptualization. **Ran Zhu:** Writing – review & editing, Software, Methodology, Investigation, Funding acquisition. **Yixuan Wu:** Writing – review & editing, Software, Resources, Methodology. **Feng Yang:** Writing – review & editing, Investigation, Funding acquisition. **Yining Cao:** Writing – review & editing, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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